

California Regional Water Quality Control Board
Santa Ana Region

March 2, 2007

ITEM: *8

SUBJECT: Renewal of Waste Discharge and Producer/User Reclamation
Requirements for the City of Rialto's Municipal Wastewater Treatment
Plant, Order No. R8-2007-0006, NPDES No. CA0105295, San Bernardino
County

DISCUSSION:

See attached Fact Sheet

RECOMMENDATIONS:

Adopt Order No. R8-2007-0006, NPDES No. CA0105295 as presented.

COMMENT SOLICITATION:

Comments were solicited from the discharger and the following agencies:

U.S. Environmental Protection Agency, Permits Issuance Section (WTR-5) – Doug Eberhardt
U.S. Army District, Los Angeles, Corps of Engineers - Regulatory Branch
U.S. Fish and Wildlife Service, Carlsbad
State Water Resources Control Board, Office of the Chief Counsel – Erik Spiess
State Department of Fish and Game, Ontario
California Coastal Conservancy – Mary Small
California Coastal Commission – Carl Schwing
California Department of Health Services, San Bernardino – Sean McCarthy
California Department of Health Services, Carpinteria - Jeff Stone
State Department of Water Resources - Glendale
San Bernardino County Department of Public Works, Environmental Management Division
San Bernardino County Environmental Health Services – Dan Avera
Santa Ana Watershed Project Authority – Celeste Cantu
Santa Ana River Dischargers Association -
Inland Empire Waterkeeper - Mandy Revell
Orange County Water District - Nira Yamachika
DDB Engineering, Inc. - Debra Burris, P.E., DEE
Orange County Coastkeeper - Garry Brown
Lawyers for Clean Water C/c San Francisco Baykeeper
City of Rialto – Derek Wieske

California Regional Water Quality Control Board

Region 8, Santa Ana- Region

3737 Main Street, Suite 500, Riverside, California 92501-3348
Phone (951) 782-4130 - FAX (951) 781-6288- TDD (951) 782-3221

<http://www.waterboards.ca.gov/santaana>

ORDER NO. R8-2007-0006

NPDES NO. CA0105295

WASTE DISCHARGE AND PRODUCER /USER RECLAMATION REQUIREMENTS FOR THE CITY OF RIALTO MUNICIPAL WASTEWATER TREATMENT PLANT DISCHARGE TO REACH 4 OF SANTA ANA RIVER

The following Discharger is subject to waste discharge requirements as set forth in this Order:

Table 1. Discharger Information

Discharger	City of Rialto
Name of Facility	City of Rialto Municipal Wastewater Treatment Plant
Address	501 E. Santa Ana Avenue
	Bloomington, CA 92316
	San Bernardino County
The U.S. Environmental Protection Agency (USEPA) and the Regional Water Quality Control Board have classified this discharge as a major discharge.	

The discharge by the City of Rialto from the discharge points identified below is subject to waste discharge requirements as set forth in this Order:

Table 2. Discharge Location

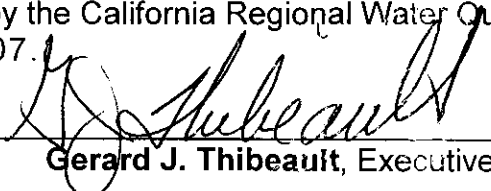
Discharge Point	Effluent Description	Discharge Point (Latitude)	Discharge Point (Longitude)	Receiving Water
001	Tertiary treated and disinfected recycled water	34°03'04"N	117°21'11"W	Lined flood control channel tributary to Santa Ana River, Reach 4, which overlies the Riverside-A Groundwater Management Zone
002	Tertiary treated and disinfected, or secondary treated and disinfected	34°02'36"N	117°21'30"W	Emergency discharge point to the Santa Ana River (Reach 4), which overlies the Riverside-A Groundwater Management Zone
003	Tertiary treated and disinfected for irrigation	34°03'12"N	117°21'36"W	Riverside-B Groundwater Management Zone; Chino 3 Groundwater Management Zone

Table 3. Administrative Information

This Order was adopted by the Regional Water Board on:	March 2, 2007
This Order shall become effective on:	March 2, 2007
This Order shall expire on:	March 2, 2012
The Discharger shall file a Report of Waste Discharge in accordance with Title 23, California Code of Regulations, as application for issuance of new waste discharge requirements no later than:	September 4, 2011

IT IS HEREBY ORDERED, that this Order supercedes Order No. 01-4 except for enforcement purposes, and, in order to meet the provisions contained in Division 7 of the California Water Code (CWC) (commencing with section 13000) and regulations adopted thereunder, and the provisions of the federal Clean Water Act (CWA) and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order.

I, Gerard J. Thibeault, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Santa Ana Region, on March 2, 2007.



Gerard J. Thibeault, Executive Officer

TABLE OF CONTENTS

I.	Facility Information.....	5
II.	Findings	5
III.	Discharge Prohibitions	12
IV.	Effluent Limitations and Discharge Specifications.....	12
	A. Effluent Limitations – Discharge Points 001 and 002	12
	1. Final Effluent Limitations – Discharge Points 001 and 002.....	12
	2. Interim Effluent Limitations - DPs 001 and 002.....	14
	3. Toxicity Requirements - DPs 001 and 002	15
	4. Effluent Limitations-DPs 001 & 002 Under Conditions of 20:1 or More Dilution	15
	B. Land Discharge Specifications - Not Applicable.	16
	C. Reclamation Specifications - DP 003	16
V.	Receiving Water Limitations and specifications	21
	A. Surface Water Limitations.....	21
	B. Groundwater Limitations.....	22
VI.	Provisions	23
	A. Standard Provisions	23
	B. Monitoring and Reporting Program Requirements (MRP)	25
	C. Special Provisions	25
	1. Reopener Provisions	25
	2. Special Studies, Technical Reports and Additional Monitoring Requirements ..	26
	3. Best Management Practices and Pollution Prevention	28
	4. Construction, Operation and Maintenance Specifications	28
	5. Special Provisions for Municipal Facilities (POTWs Only)	29
	6. Other Special Provision - Not Applicable	32
	7. Compliance Schedules – DP 001 and DP 002	32
VII.	Compliance Determination.....	33

LIST OF TABLES

Table 1. Discharger Information.....	1
Table 2. Discharge Location	1
Table 3. Administrative Information.....	2
Table 4. Facility Information	5
Table 5. Basin Plan Beneficial Uses	8
Table 6. Effluent Limitations at DPs 001 & 002.....	13
Table 7. Interim Effluent Limitations at DPs 001 & 002.....	14
Table 8. Effluent Limitations at DPs 001 & 002 Under 20:1 Dilution	15
Table 9. Recycled Water Effluent Limitations.....	16
Table 10. Recycled Water TDS Limitations.....	17
Table 11. Recycled Water TIN Limitations.....	17

LIST OF ATTACHMENTS

Attachment A – Definitions.....	A-1
Attachment B – Location Map	B-1
Attachment C – Flow Schematic.....	C-1
Attachment D – Standard Provisions	D-1
Attachment E – Monitoring and Reporting Program (MRP)	E-1
Attachment F – Fact Sheet	F-1
Attachment G – EPA PPL Table	G-1
Attachment H – ML Table	H-1
Attachment I – Triggers for Monitoring Priority Pollutants	I-1

I. FACILITY INFORMATION

The following Discharger is subject to waste discharge requirements as set forth in this Order:

Table 4. Facility Information

Discharger	City of Rialto
Name of Facility	Municipal Wastewater Treatment Plant
Address	501 E. Santa Ana Ave.
	Bloomington, CA 92316
	San Bernardino County
Facility Contact, Title and Phone	David Kachelski,, Project Manager, Veolia Water North America West, LLC, City of Rialto, Phone: (909) 877-2752
Mailing Address	501 E. Santa Ana Ave., Bloomington, CA 92316
Type of Facility	POTW
Facility Design Flow	11.7 million gallons per day tertiary treatment

II. FINDINGS

The California Regional Water Quality Control Board, Santa Ana Region (hereinafter Regional Water Board), finds:

A. Background. The City of Rialto (hereinafter Discharger) is currently discharging pursuant to Order No. 01-4 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0105295. The Discharger submitted a Report of Waste Discharge, dated December 12, 2005, and applied for a NPDES permit renewal to discharge up to 11.7 million gallons per day (mgd) of tertiary treated wastewater from its Rialto Municipal Wastewater Treatment Plant, hereinafter Facility. The application was deemed complete on December 20, 2006.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

B. Facility Description. The Discharger owns the Rialto Municipal Wastewater Treatment Plant (RMWTP). Veolia Water North America West, LLC is a private contractor operating the Facility for the City. The treatment systems consist of primary, secondary, and tertiary wastewater treatment. Wastewater from this Facility is discharged at Discharge Point 001 or at emergency Discharge Point 002 (see Table on cover page) to Reach 4 of the Santa Ana River, which is a water of the United States. Recycled water is presently delivered from Discharge Point 003 for irrigation and groundwater recharge within the Riverside-B and Chino 3 Groundwater Management Zones. Attachment B provides a vicinity map of the facility. Attachment C provides a flow schematic of the treatment system at this Facility.

- C. Legal Authorities.** This Order is issued pursuant to Section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (CWC) (commencing with section 13370). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4, Division 7 of the Water Code (commencing with section 13260).
- D. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. Attachment F, which contains background information and rationale for Order requirements, is hereby incorporated into this Order and, thus constitutes part of the Findings for this Order. Attachments A through E, and G through I are also incorporated into this Order.
- E. State law.** The provisions/requirements in subsections IV.C. and V.B. of this Order are included to implement State law only. These provisions are not required or authorized under the federal CWA; consequently, violations of these provisions are not subject to the enforcement remedies that are available for NPDES violations.
- F. California Environmental Quality Act (CEQA).** Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100 et seq. (*County of Los Angeles v. California State Water Resources Control Board* (2006) 143 Cal.App.4th 985, mod. (Nov. 6, 2006, B184034) 50 Cal.Rptr.3d 619, 632-636.) This action also involves the re-issuance of waste discharge requirements for an existing facility that discharges treated wastewater to land and as such, is exempt from the provisions of California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301.
- G. Technology-based Effluent Limitations.** Section 301(b) of the CWA and implementing USEPA permit regulations at section 122.44, title 40 of the Code of Federal Regulations¹, require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at Part 133 and/or Best Professional Judgment (BPJ) in accordance with Part 125, section 125.3. A detailed discussion of the technology-based effluent limitations development is included in the Fact Sheet.
- H. Water Quality-based Effluent Limitations.** Section 301(b) of the CWA and section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of tertiary treatment requirements, is discussed in the Fact Sheet.

¹ All further statutory references are to title 40 of the Code of Federal Regulations unless otherwise indicated.

Section 122.44(d)(1)(i) mandates that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

- I. **Water Quality Control Plans.** The Regional Water Board adopted a revised Water Quality Control Plan for the Santa Ana Region (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters in the Santa Ana Region addressed through the plan. More recently, the Basin Plan was amended significantly to incorporate revised boundaries for groundwater subbasins, now termed "management zones", new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. This Basin Plan Amendment was adopted by the Regional Water Board on January 22, 2004. The State Water Resources Control Board (State Water Board) and Office of Administrative Law (OAL) approved the Amendment on September 30, 2004 and December 23, 2004, respectively. The surface water standards provisions of the Amendment are awaiting approval by the U.S. Environmental Protection Agency but have no bearing on the requirements of this Order. This Order implements applicable provisions of the N/TDS Basin Plan amendment, which, for the Discharger, results in effluent limitations that are at least as stringent as those in the current permit.

In addition, the Basin Plan implements State Water Board Resolution No. 88-63, which established State Policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Based on the criteria specified in the State Water Board Resolution, the Basin Plan specifies that the Santa Ana River, Reach 4 and downstream reaches of the River are excepted from the municipal and domestic supply beneficial use. As discussed in detail in the Fact Sheet (Attachment F), beneficial uses applicable to Reach 4 of the Santa Ana River are as follows:

Table 5. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 002	Reach 4 of Santa Ana River	<u>Present or Potential:</u> Groundwater recharge (GWR), water contact recreation (REC-1), non-contact water recreation (REC-2), warm freshwater habitat, and wildlife habitat (WILD) Excepted from Municipal and Domestic supply
001, 002	Riverside-A Management Zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Riverside-B Management Zone and Chino 3 "antidegradation" management zone ²	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
001, 002, 003	Downgradient groundwater management zones	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.

Requirements of this Order implement the Basin Plan.

- J. National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain water quality criteria for priority pollutants.

² The Chino 3 "antidegradation" Management Zone is a part of the Chino North "maximum benefit" Management Zone to which maximum benefit objectives for total dissolved solids and nitrate- nitrogen apply if the Chino Basin Watermaster (Watermaster) and the Inland Empire Utilities District (IEUA) fulfill a maximum benefit program specified in the Basin Plan. These maximum benefit TDS and nitrate-nitrogen objectives apply for the purpose of regulating the Discharger, in lieu of the Chino 3 "antidegradation" objectives for these constituents, provided that the Watermaster/IEUA confirm that discharges by the Discharger are in conformance with their maximum benefit program.

- K. State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP became effective on April 28, 2000 with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000 with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005 that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
- L. State General Waste Discharge Requirements for Sanitary Sewer Systems.** The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order No. 2006-0003 on May 2, 2006, requiring public agencies that own sanitary sewer systems, comprised of more than one mile of pipes or sewer lines, to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs).

This Order requires the Discharger and other government agencies³ to obtain enrollment for regulation under the General Water Quality Order No. 2006-0003.

- M. Compliance Schedules and Interim Requirements.** Section 2.1 of the SIP provides that, based on a Discharger's request and demonstration that it is infeasible for an existing Discharger to achieve immediate compliance with an effluent limitation derived from a CTR criterion, compliance schedules may be allowed in an NPDES permit. Unless an exception has been granted under Section 5.3 of the SIP, a compliance schedule may not exceed 5 years from the date that the permit is issued or reissued, nor may it extend beyond 10 years from the effective date of the SIP (or May 18, 2010) to establish and comply with CTR criterion-based effluent limitations. Where a compliance schedule for a final effluent limitation exceeds 1 year, the Order must include interim numeric limitations for that constituent or parameter. Where allowed by the Basin Plan, compliance schedules and interim effluent limitations or discharge specifications may also be granted to allow time to implement a new or revised water quality objective. This Order includes compliance schedules and interim effluent limitations and/or discharge specifications. A detailed discussion of the basis for the compliance schedule(s) and interim effluent limitation(s) and/or discharge specifications is included in the Fact Sheet.

³ Member agencies and sewerage agencies discharging wastewater into the facility.

- N. Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised State and Tribal water quality standards (WQS) become effective for CWA purposes (40 CFR 131.21, 65 FR 24641, April 27, 2000). Under the revised regulation (also known as the Alaska rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.
- O. Stringency of Requirements for Individual Pollutants.** This Order contains restrictions on individual pollutants that are no more stringent than required by the federal CWA. Individual pollutant restrictions consist of technology-based restrictions and water quality-based effluent limitations. The technology-based effluent limitations consist of restrictions on pollutants listed. Restrictions on pollutants listed are specified in federal regulations as discussed in Attachment F, and the permit's technology-based pollutant restrictions are no more stringent than required by the CWA. Water quality-based effluent limitations have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant water quality-based effluent limitations were derived from the CTR, the CTR is the applicable standard pursuant to 40 CFR 131.38. The scientific procedures for calculating the individual water quality-based effluent limitations are based on the CTR-SIP, which was approved by USEPA on May 18, 2000. Apart from certain surface water standards changes resulting from the N/TDS Basin Plan amendment that do not materially affect the quality requirements for the discharges regulated by this Order, all beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the technology-based requirements of the CWA and the applicable water quality standards for purposes of the CWA.
- P. Antidegradation Policy.** Section 131.12 of 40 CFR requires that State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy, where the federal policy applies under federal law. Resolution 68-16 requires that existing quality of waters be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. As discussed in detail in the Fact Sheet (Attachment F) the permitted discharge is consistent with the antidegradation provision of 40 CFR section 131.12 and State Water Board Resolution 68-16.

- Q. Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and federal regulations at title 40 Code of Federal Regulations, section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order No. 01-4.
- R. Biosolids Requirements.** On February 19, 1993, the USEPA issued a final rule for the use and disposal of sewage sludge, 40 CFR, Part 503. This rule requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. The State of California has not been delegated the authority to implement this program, therefore, the U.S. Environmental Protection Agency is the implementing agency. However, this Order includes Regional Board biosolids requirements.
- S. Monitoring and Reporting.** Section 122.48 of 40 CFR requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorize the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. This Monitoring and Reporting Program is provided in Attachment E.
- T. Standard and Special Provisions.** Standard Provisions, which apply to all NPDES permits in accordance with section 122.41, and additional conditions applicable to specified categories of permits in accordance with section 122.42, are provided in Attachment D. The Discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42. The Regional Water Board has also included in this Order special provisions applicable to the Discharger. A rationale for the special provisions contained in this Order is provided in the attached Fact Sheet.
- U. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe Waste Discharge Requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of notification are provided in the Fact Sheet (Attachment F) of this Order.
- V. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet (Attachment F) of this Order.

III. DISCHARGE PROHIBITIONS

- A. Wastewater discharged shall be limited to treated and disinfected effluent that meets the conditions and requirements specified in Section IV.A.1., except for discharges of treated wastewater that meets the conditions specified in Section IV.A.4. when the flow⁴ in the River results in a dilution of 20:1 or more at the point of discharge.
- B. The direct discharge of secondary treated wastewater to Reach 4 of Santa Ana River other than when the flow⁴ in the River results in a dilution of 20:1 or more at the point of discharge is prohibited.
- C. The Discharge of wastewater at a location or in a manner different from those described in this Order is prohibited.
- D. The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited, excepted as allowed in Standard Provisions - Permit Compliance I. G. of Attachment D, Standard Provisions.
- E. The discharge of any substances in concentrations toxic to animal or plant life is prohibited.
- F. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations – Discharge Points 001 and 002

1. Final Effluent Limitations – Discharge Points 001 and 002

Unless otherwise specifically specified hereinafter, compliance with the following effluent limitations is measured at monitoring locations M-001 and M-002 as described in the attached MRP (Attachment E)

- a. The Discharge shall maintain compliance with the following effluent limitations at DP 001 and DP 002,:

⁴ *Exclusive of discharges to surface waters from upstream publicly owned treatment works.*

Table 6. Effluent Limitations at DPs 001 & 002

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	--	--	--
Total Suspended Solids	mg/L	20	30	--	--	--
pH ⁵	standard units	--	--	--	6.5	8.5
Total Chlorine Residual	mg/L	--	--	--	--	0.1
Ammonia-Nitrogen	mg/L	4.5	--	--	--	--
Total Recoverable Copper	µg/L	26.1	--	52.4	--	--
Cyanide, Free*	µg/L	4.2	--	8.5	--	--
Total Recoverable Mercury*	µg/L	0.051	--	0.102	--	--
Bis(2-ethylhexyl) phthalate*	µg/L	5.9	--	11.8	--	--

* Final Limits specified herein are effective starting on March 2, 2008 and thereafter. Interim effluent limitations are specified for this constituent in IV.A.2., below.

- b. Percent Removal: The average monthly percent removal of BOD 5-day 20°C and total suspended solids shall not be less than 85 percent.
- c. TDS Limitations: The lower of the two total dissolved solids limits specified in 1) or 2), below, is the limit.
 - 1) The 12-month flow weighted running average total dissolved solids concentration and mass emission rate shall not exceed 490 mg/L and 49,039 lbs/day⁶, respectively.
 - 2) The 12-month flow weighted running average total dissolved solids concentration shall not exceed the 12-month flow weighted average total dissolved solids concentration in the water supply by more than 250 mg/L⁷.

⁵ See Section VII.K. – Compliance Determination.

⁶ Derived from 12 mgd x 8.34 x 490 mg/L.

⁷ An exceedance of this limit shall not be considered a violation, provided it is due solely to chemical additions in the treatment process needed to meet waste discharge requirements or other valid regulatory requirements. TDS quality is used to evaluate the water supply mineral increment may be measured at influent or secondary effluent.

- d. Total Inorganic Nitrogen (TIN) Limitations: The 12-month flow weighted running average TIN concentration and mass emission rate of the discharge shall not exceed 10 mg/L and 1,001 lbs/day⁸, respectively.
- e. The discharge shall at all times be adequately oxidized, filtered, and disinfected treated wastewater and shall meet the following limitations.
 - 1) When filtration is through natural undisturbed soils or a bed of filter media, the turbidity of the filter effluent shall not exceed any of the following:
 - a) Average of 2 Nephelometric Turbidity Unit (NTU) within any calendar day;
 - b) 5 NTU more than 5 percent of the time in any calendar day; and
 - c) 10 NTU at any time.
 - 2) When chlorine disinfection process is utilized followed by filtration, a CT (the product of total chlorine residual and modal contact time measured at the same point) value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow shall be provided.
 - 3) Where ultraviolet (UV) disinfection is used for disinfection, UV disinfection shall meet the requirements specified in the Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse, published by the National Water Research Institute, Second Edition, unless otherwise approved by the California Department of Health Services.
 - 4) The weekly average concentration of total coliform bacteria shall not exceed a Most Probable Number (MPN) of 2.2 total coliform bacteria per 100 milliliters (ml). (see Compliance Determination VII.J.1., below)
 - 5) The number of total coliform bacteria shall not exceed an MPN of 23 total coliform bacteria per 100 ml in more than one sample in any calendar month.
 - 6) No total coliform bacteria sample shall exceed an MPN of 240 total coliform bacteria per 100 ml.
- f. There shall be no visible oil and grease in the discharge.

2. Interim Effluent Limitations - DPs 001 and 002

During the period beginning March 2, 2007 and ending on March 1, 2008, the discharge of tertiary effluent shall maintain compliance with the following limitations at Discharge Points 001 and 002, with compliance measured at Monitoring Locations M-001 and M-002, respectively, as described in the attached MRP.

Table 7. Interim Effluent Limitations at DPs 001 & 002

Parameter	Units	Average Monthly	Maximum Daily
Cyanide, Free	µg/L	24	24

⁸ Derived from 12 mgd x 8.34 x 10 mg/L.

Table 7. Interim Effluent Limitations at DPs 001 & 002

Parameter	Units	Average Monthly	Maximum Daily
Total Recoverable Mercury	µg/L	0.23	0.23
Bis(2-ethylhexyl) phthalate	µg/L	15	15

3. Toxicity Requirements - DPs 001 and 002

Compliance with toxicity requirements shall be measured at monitoring location M-001 and M-002:

- There shall be no acute or chronic toxicity in the plant effluent nor shall the plant effluent cause any acute or chronic toxicity in the receiving water. All waters shall be maintained free of toxic substances in concentrations which are toxic to, or which produce detrimental physiological responses in human, plant, animal, or indigenous aquatic life. This Order contains no numeric limitation for toxicity. However, the Discharger shall conduct chronic toxicity monitoring.
- The Discharger shall implement the accelerated monitoring as specified in Attachment E when the result of any single chronic toxicity test of the effluent exceeds 1.0 TUc.

4. Effluent Limitations-DPs 001 & 002 Under Conditions of 20:1 or More Dilution

- The discharge of treated and disinfected effluent when the river flow⁹ at monitoring location R-001U in Santa Ana River results in a dilution of 20:1 (receiving water flow:wastewater flow) or more at DP 001 and DP 002 shall maintain compliance with the following effluent limitations, with compliance measured at monitoring location M-001 and M-002, respectively as described in the attached MRP (Attachment E):

Table 8. Effluent Limitations at DPs 001 & 002 Under 20:1 Dilution

Parameter	Units	Effluent Limitations			
		Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	30	45	--	--
Total Suspended Solids	mg/L	30	45	--	--
pH	standard units	--	--	6.5	8.5
Total Residual Chlorine	mg/L	--	--	--	2.1 *

* See Attachment F, p. F-18

⁹

Exclusive of discharges to surface waters from upstream publicly owned treatment works.

- b. Treated wastewater shall at all times be adequately oxidized and disinfected wastewater and shall meet the following limitations, with compliance measured at monitoring location M-001 and M-002:
 - 1) The weekly average number of coliform bacteria does not exceed a median of 23 per 100 milliliters as determined from the daily coliform bacteria values for the last seven (7) days. (see also Compliance Determination VII.J.2., below)
 - 2) The discharge shall be considered adequately oxidized if the 5-day @ 20°C Biochemical Oxygen Demand and Total Suspended Solids constituent concentrations of the discharge are less than or equal to the limitations shown in IV.A.4.a., above.
- c. The monthly average biochemical oxygen demand and suspended solids concentrations of the discharge shall not be greater than fifteen percent (15%) of the monthly average influent concentration.

B. Land Discharge Specifications - Not Applicable.

C. Reclamation Specifications - DP 003

1. The use of recycled water for landscape irrigation or other similar uses shall maintain compliance with the following limitations. Compliance is to be measured at representative monitoring location REC-001 or other approved monitoring locations where representative samples of recycled water can be obtained for laboratory testing and analysis, as described in the attached MRP (Attachment E). The Discharger shall submit for approval by the Executive Officer other monitoring location(s) not specified herein where representative samples of recycled water could be obtained for laboratory testing and analysis.

- a. Physical/Biological Limitations with compliance measured at monitoring location:

Table 9. Recycled Water Effluent Limitations

Parameter	Units	Effluent Limitations			
		Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	--	--
Total Suspended Solids	mg/L	20	30	--	--
pH	standard units	--	--	6	9

- b. TDS Limitations: The following TDS limitations apply to the recycled water to be used for irrigation on sites overlying the specific Groundwater Management Zones identified herein:

Table 10. Recycled Water TDS Limitations

Groundwater Management Zone	12-month flow weighted running average TDS limitations, mg/L
Riverside A	560
Riverside-B	290
Chino 3—Anti-degradation (see IV.C.1.b.1, below)	260
Chino North—Max benefit, (see IV.C.1.b.2, below)	420

- 1) The 260 mg/L TDS limit for recycled water use overlying the Chino 3 groundwater management zone applies unless the Discharger submits to the Regional Board confirmation from Chino Basin Watermaster/Inland Empire Utilities Agency (CBW/IEUA) that the Discharger's proposed recycling project within the Chino 3 Groundwater Management Zone is consistent with CBW/IEUA's maximum benefit program (see Provisions VI.C.2.c.).
 - 2) The 420 mg/L TDS limit for recycled water use overlying the Chino North groundwater management zone applies when the Discharger submits to the Regional Board confirmation from CBW/IEUA that the Discharger's proposed recycling project is consistent with CBW/IEUA's maximum benefit program (see Provisions VI.C.2.c.).
- c. Total Inorganic Nitrogen (TIN) Limitations: The following TIN limitations apply to recycled water use sites overlying the specific Groundwater Management Zone identified herein:

Table 11. Recycled Water TIN Limitations

Groundwater Management Zone	12-month flow weighted running average TIN limitations, mg/L
Riverside A	8.3
Riverside-B	10.1
Chino 3—Anti-degradation (see IV.C.1.c.1, below)	4.7
Chino North—Max benefit (see IV.C.1.c.2, below)	6.7

- 1) The 4.7 mg/L TIN limit applies to recycled water use overlying the Chino 3 groundwater management zone unless the Discharger submits to the Regional Board confirmation from CBW/IEUA that the Discharger's proposed recycling project within the Chino 3 Groundwater Management Zone is consistent with CBW/IEUA's maximum benefit program (see Provisions VI.C.2.c.).

- 2) The 6.7 mg/L TIN limit for recycled water use overlying the Chino North applies when the Discharger submits to the Regional Board confirmation from CBW/IEUA that the Discharger's proposed recycling project is consistent with CBW/IEUA's maximum benefit program (see Provisions VI.C.2.c.).
- d. Recycled water described in Section 60307(a) of Division 4, Chapter 3, Title 22, California Code of Regulations and for irrigation of food crops, parks and playground, school yards, residential landscaping and other irrigation uses not specified in Section 60304(a) of Division 4, Chapter 3, Title 22, California Code of Regulations or not prohibited in other Sections of the California Code of Regulations shall at all times be adequately oxidized, filtered, and disinfected tertiary treated wastewater and shall meet the following limitations:
 - 1). The turbidity of the filter effluent shall not exceed any of the following:
 - a). When filtration is through natural undisturbed soils or a bed of filter media:
 - (1) Average of 2 Nephelometric Turbidity Units (NTU) within any calendar day;
 - (2) 5 NTU more than 5 percent of the time in any calendar day; and
 - (3) 10 NTU at any time.
 - 2). The disinfected effluent shall meet the following:
 - a) The weekly average total coliform bacteria¹⁰ shall not exceed a Most Probable Number (MPN) of 2.2 total coliform bacteria per 100 milliliters (ml).
 - b) The number of total coliform organism shall not exceed an MPN of 23 total coliform bacteria per 100 ml in more than one sample in any calendar month.
 - c) No total coliform sample shall exceed an MPN of 240 total coliform bacteria per 100 ml.
 - d) When chlorine disinfection process is utilized followed by filtration, a CT (the product of total chlorine residual and modal contact time¹¹ measured at the same point) value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow shall be provided.
 - e) When a disinfection process combined with the filtration process is utilized, the combined process shall demonstrate inactivation and/or removal of 99.999 percent of the plaque-forming units of F-specific bacteriophage MS2, or polio virus in the wastewater. A virus that is at least as resistant to disinfection as polio virus may be used for purposes of the demonstration.

¹⁰ See Compliance Determination Section VII.J.1.

¹¹ Modal contact time shall be calculated daily based on the minimum one-hour average value in a 24-hour period.

- f) Where ultraviolet (UV) disinfection is used for disinfection, UV disinfection shall meet the requirements specified in the Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse, published by the National Water Research Institute, Second Edition, unless otherwise approved by the California Department of Health Services
- e. Recycled water used for irrigation of school yards, parks, golf courses with unrestricted access, landscape, and food crops where the edible portion is produced above ground and not contacted by the recycled water shall at all times be adequately oxidized and disinfected so that average weekly total coliform bacteria in the disinfected effluent does not exceed a most probable number (MPN) of 2.2 per 100 milliliters utilizing the bacteriological results of the last seven days for which analyses have been completed, and the number of total coliform bacteria does not exceed an MPN of 23 per 100 milliliters in more than one sample in any calendar month.
- f. Recycled water used for the uses listed below shall be an oxidized and disinfected water so that average weekly total coliform bacteria¹² in the disinfected effluent does not exceed a most probable number (MPN) of 23 per 100 milliliters utilizing the bacteriological results of the last seven days for which analyses have been completed, and the number of total coliform bacteria does not exceed an MPN of 240 per 100 milliliters in more than one sample in any 30 day period.
 - 1) Industrial boiler feed, nonstructural fire fighting, backfill consolidation around nonpotable piping, soil compaction, mixing concrete, dust control on roads and streets, cleaning roads, sidewalks and outdoor work areas and industrial process water that will not come into contact with workers.
 - 2) Irrigation of cemeteries, freeway landscaping, restricted access golf courses, ornamental nursery stock and sod farms where access by the general public is not restricted, pasture for animals producing milk for human consumption, and any nonedible vegetation where access is controlled so that irrigated area cannot be used as if it were part of a park, playground or school yard.
- g. For recycled water uses specified in Sections 60304 and 60307 of Title 22 where filtration is provided pursuant Section 60301.320(a) and coagulation is not used as part of the treatment process, the Discharger shall comply with the following:
 - 1) The turbidity of the influent to the filters is continuously measured and the influent turbidity does not exceed 5 NTU for more than 15 minutes and never exceeds 10 NTU;
 - 2) The filter effluent turbidity shall not exceed 2 NTU; and;

¹² See Compliance Determination Section VII.J.2.

- 3) Should the filter influent turbidity exceed 5 NTU for more than 15 minutes, chemical addition shall be automatically activated if available, if not, the wastewater shall be diverted.
2. For new reuse sites, the use of recycled water shall only commence after the California Department of Health Services (CDHS) grants final approval for such use. The Discharger shall provide the Regional Water Board with a copy of the CDHS approval letter within 30 days of the approval notice.
3. The Discharger shall be responsible for assuring that recycled water is delivered and utilized in conformance with this Order, the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations. The Discharger shall conduct periodic inspections of the facilities of the recycled water users to monitor compliance by the users with this Order.
4. The Discharger shall establish and enforce Rules and Regulations for Recycled Water users, governing the design and construction of recycled water use facilities and the use of recycled water in accordance with the uniform statewide recycling criteria established pursuant to the California Water Code Section 13521.
 - a. Use of recycled water by the Discharger shall be consistent with its Rules and Regulations for Recycled Water Use.
 - b. Any revisions made to the Rules and Regulations shall be subject to the review of the Regional Water Board, the California Department of Health Services, and the County Environmental Health Department. The revised Rules and Regulations or a letter certifying that the Discharger's Rules and Regulations contain the updated provisions in this Order, shall be submitted to the Regional Water Board within 60 days of adoption of this Order by the Regional Water Board.
5. The Discharger shall, within 60 days of the adoption of this Order, review and update as necessary its program to conduct compliance inspections of recycled water reuse sites. Inspections shall determine the status of compliance with the Discharger's Rules and Regulations for Recycled Water Use.
6. The storage, delivery, or use of recycled water shall not individually or collectively, directly or indirectly, result in a pollution or nuisance, or adversely affect water quality, as defined in the California Water Code.
7. Prior to delivering recycled water to any new user, the Discharger shall submit to the Regional Water Board, the California Department of Health Services and the County Environmental Health Department a report containing the following information for review and approval:
 - a. The average number of persons estimated to be served at each use site area on a daily basis.

- b. The specific boundaries of the proposed use site area including a map showing the location of each facility, drinking water fountain, and impoundment to be used.
 - c. The person or persons responsible for operation of the recycled water system at each use area.
 - d. The specific use to be made of the recycled water at each use area.
 - e. The methods to be used to assure that the installation and operation of the recycled system will not result in cross connections between the recycled water and potable water piping systems. This shall include a description of the pressure, dye or other test methods to be used to test the system.
 - f. Plans and specifications which include following:
 - 1) Proposed piping system to be used.
 - 2) Pipe locations of both the recycled and potable systems.
 - 3) Type and location of the outlets and plumbing fixtures that will be accessible to the public.
 - 4) The methods and devices to be used to prevent backflow of recycled water into the potable water system.
 - 5) Plan notes relating to specific installation and use requirements.
8. The Discharger shall require the user(s) to designate an on-site supervisor responsible for the operation of the recycled water distribution system within the recycled water use area. The supervisor shall be responsible for enforcing this Order, prevention of potential hazards, the installation, operation and maintenance of the distribution system, maintenance of the distribution and irrigation system plans in "as-built" form, and for the distribution of the recycled wastewater in accordance with this Order.

V. RECEIVING WATER LIMITATIONS AND SPECIFICATIONS

A. Surface Water Limitations

1. Receiving water limitations are based upon water quality objectives contained in the Basin Plan. As such, they are a required part of this Order. The discharge shall not cause the following in downstream Reaches of the Santa Ana River:
 - a. Coloration of the receiving waters, which causes a nuisance or adversely affects beneficial uses.
 - b. Deposition of oil, grease, wax or other materials in the receiving waters in concentrations which result in a visible film or in coating objects in the water, or which cause a nuisance or affect beneficial uses.
 - c. An increase in the amounts of suspended or settleable solids in the receiving waters, which will cause a nuisance or adversely affect beneficial uses as a result of controllable water quality factors.
 - d. Taste or odor-producing substances in the receiving waters at concentrations, which cause a nuisance or adversely affect beneficial uses.

- e. The presence of radioactive materials in the receiving waters in concentrations, which are deleterious to human, plant or animal life.
 - f. The depletion of the dissolved oxygen concentration below 5.0 mg/L.
 - g. The temperature of the receiving waters to be raised above 90°F (32°C) during the period of June through October, or above 78°F (26°C) during the rest of the year.
 - h. The concentration of pollutants in the water column, sediments, or biota to adversely affect the beneficial uses of the receiving water. The discharge shall not result in the degradation of inland surface water communities and populations, including vertebrate, invertebrate, and plant species.
- 2. The discharge of wastes shall not cause a violation of any applicable water quality standards for receiving waters adopted by the Regional Water Board or State Water Board, as required by the Clean Water Act and regulations adopted thereunder.
 - 3. Pollutants not specifically mentioned and limited in this Order shall not be discharged at levels that will bioaccumulate in aquatic resources to levels, which are harmful to human health.
 - 4. The discharge shall not contain constituent concentrations of mercury that will result in the bioaccumulation of methylmercury in fish flesh tissue greater than 0.3 milligram methylmercury/kilogram. (See also Section VI.C.1.e. and VI.C.2.a, below).

B. Groundwater Limitations

- 1. The use of recycled water shall not cause the underlying groundwater to be degraded, to exceed water quality objectives, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

VI. PROVISIONS

A. Standard Provisions

1. The Discharger shall comply with all Standard Provisions included in Attachment D of this Order.
2. Regional Water Board Standard Provisions. The Discharger shall comply with the following provisions:
 - a. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
 - b. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, discharge limitations (e.g., maximum daily effluent limitation), or receiving water limitation of this Order, the Discharger shall notify the Regional Water Board by telephone (951) 782-4130 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within five days, unless the Regional Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and, prevent recurrence including, where applicable, a schedule of implementation. Other noncompliance requires written notification as above at the time of the normal monitoring report.
 - c. Neither the treatment nor the discharge of pollutants shall create a pollution, contamination, or nuisance as defined by Section 13050 of the CWC.
 - d. The Discharger shall take all reasonable steps to minimize or correct any adverse impact on the environment resulting from noncompliance with this Order, including such accelerated or additional monitoring as may be necessary to determine the nature and impact of the noncomplying discharge.
 - e. This Order may be modified, revoked and reissued, or terminated for cause including, but not limited to, the following.
 - 1). Violation of any terms or conditions of this Order;
 - 2). Obtaining this Order by misrepresentation or failure to disclose fully all relevant facts, or;

- f. In addition to any other grounds specified herein, this permit may be modified or revoked at any time if, on the basis of any data, the Regional Water Board determines that continued discharges may cause unreasonable degradation of the aquatic environment.
- g. If an effluent standard or discharge prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is established under Section 307 (a) of the Clean Water Act for a toxic pollutant which is present in the discharge, and such standard or prohibition is more stringent than any limitation for that pollutant in this Order, this Order may be modified or revoked and reissued to conform to the effluent standard or discharge prohibition.
- h. The discharger shall file with the Regional Water Board a Report of Waste Discharge at least 180 days before making any material change in the character, location, or volume of the discharge. A material change includes, but is not limited to, the following:
 - 1). Adding a major industrial waste discharge to a discharge of essentially domestic sewage, or adding a new process or product by an industrial facility resulting in a change in the character of the waste.
 - 2). Significantly changing the disposal method or location, such as changing the disposal to another drainage area or water body.
 - 3). Significantly changing the method of treatment.
 - 4). Increasing the treatment plant design capacity beyond that specified in this Order.
- i. The provisions of this Order are severable, and if any provision of this Order, or the application of any provision of this Order to any circumstance, is held invalid, the application of such provision to other circumstances, and the remainder of this Order, shall not be affected thereby.
- j. The Discharger shall maintain a copy of this Order at the site so that it is available to site operating personnel at all times. Key operating personnel shall be familiar with its content.
- k. The Discharger shall optimize chemical additions needed in the treatment process to meet waste discharge requirements so as to minimize total dissolved solid increases in the treated wastewater.
- l. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Regional Water Board's Executive Officer.
- m. If the Discharger demonstrates a correlation between the biological oxygen demand (BOD5) and total organic carbon (TOC) concentrations in the effluent to the satisfaction of the Executive Officer, compliance with the BOD5 limits contained in this Order may be determined based on analyses of the TOC of the effluent.

- n. In the event of any change in control or ownership of land or waste discharge facility presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Regional Water Board.
- o. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

B. Monitoring and Reporting Program Requirements (MRP)

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E of this Order. This monitoring and reporting program may be modified by the Executive Officer at any time during the term of this Order, and may include an increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected. Any increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected may be reduced back to the levels specified in the original monitoring and reporting program at the discretion of the Executive Officer.

C. Special Provisions

1. Reopener Provisions

- a. This Order will be reopened to address any changes in State or federal plans, policies or regulations that would affect the quality requirements for the discharges.
- b. This Order may be reopened to include effluent limitations for pollutants determined to be present in the discharge in concentrations that pose a reasonable potential to cause or contribute to violations of water quality objectives.
- c. This Order may be reopened and modified in accordance with the requirements set forth at 40 CFR 122 and 124, to include the appropriate conditions or limits to address demonstrated effluent toxicity based on newly available information, or to implement any EPA-approved new State water quality standards applicable to effluent toxicity.
- d. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.
- e. This Order may be reopened to include an appropriate bioaccumulation based effluent limit for mercury if test results (as required in Attachment E of this Order) show that the concentration levels of methylmercury in the fish tissue are at or above 0.3 milligrams per kilogram.

- f. This Order may be reopened to incorporate appropriate biosolids requirements if the State Water Board and the Regional Water Board are given the authority to implement regulations contained in 40 CFR 503.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

- a. By June 1, 2007, the Discharger shall notify the Executive Officer of its continuous involvement with the comprehensive mercury investigation program currently being conducted by a group of Santa Ana River system Dischargers. If the Discharger discontinues its involvement with this comprehensive program, the Discharger shall, within 60 days of that date, submit for the approval of the Executive Officer its plan for the annual testing of mercury levels in fish flesh samples collected from the Santa Ana River, upstream of, at, and downstream of the point of the discharge point. Upon approval, the Discharger shall implement the plan.
- b. By June 1, 2007, the Discharger shall submit for approval by the Executive Officer, a report that details the manner in which sampling, monitoring and reporting will be performed as required in the Order.
- c. Application of the TDS and TIN limitations for recycled water use in the Chino North groundwater management zone is contingent on the submission of confirmation by CBW/IEUA that the proposed recycled water use is consistent with the maximum benefit program being implemented by these agencies (see IV.C.1.b. If application of TDS and TIN limits for recycled water use in the Chino North Management Zone is pursued by the Discharger, this confirmation shall be submitted no later than May 1, 2007. In the absence of this confirmation, the TDS and TIN limits for recycled water use in the Chino 3 groundwater management zone shall apply. In addition, the Discharger shall be responsible to mitigate TDS and TIN discharges that occurred in excess of the limits based on the Chino 3 groundwater management zone objectives during the period between issuance of this Order and May 1, 2007. A proposal for TDS and TIN mitigation, if necessary, shall be submitted by June 1, 2007 and implemented upon approval by the Executive Officer.
- d. Toxicity Reduction Requirements.
 - 1) The Discharger shall develop an Initial Investigation Toxicity Reduction Evaluation (IITRE) work plan that describes the steps the Discharger intends to follow if required by Toxicity Requirement d. 2), below. The work plan shall include at a minimum:
 - a) A description of the investigation and evaluation techniques that will be used to identify potential causes/sources of the exceedance, effluent variability, and/or efficiency of the treatment system in removing toxic substances. This shall include a description of an accelerated chronic toxicity testing program.

- b) A description of the methods to be used for investigating and maximizing in-house treatment efficiency and good housekeeping practices.
 - c) A description of the evaluation process to be used to determine if implementation of a more detailed TRE/TIE is necessary.
- 2) The Discharger shall implement the IITRE work plan whenever the results of chronic toxicity tests of the effluent exceed:
 - a) A two month median value of 1.0 TUC for survival or reproduction endpoint or,
 - b) Any single test value of 1.7 TUC for survival endpoint.
- 3) The Discharger shall develop a detailed Toxicity Reduction Evaluation and Toxicity Identification Evaluation (TRE/TIE) work plan that shall describe the steps the Discharger intends to follow if the implemented IITRE fails to identify the cause of, or to rectify, the toxicity.
- 4) The Discharger shall use as guidance, at a minimum, EPA manuals EPA/600/2-88/070 (industrial), EPA/600/4-89-001A (municipal), EPA/600/6-91/005F (Phase I), EPA/600/R-92/080 (Phase II), and EPA-600/R-92/081 (Phase III) to identify the cause(s) of toxicity. If during the life of this Order the aforementioned EPA manuals are revised or updated, the revised/updated manuals may also be used as guidance. The detailed TRE/TIE work plan shall include:
 - a) Further actions to investigate and identify the cause of toxicity;
 - b) Actions the Discharger will take to mitigate the impact of the discharge and to prevent the recurrence of toxicity; and
 - c) A schedule for these actions.
- 5) The Discharger shall implement the TRE/TIE workplan if the IITRE fails to identify the cause of, or rectify, the toxicity, or if in the opinion of the Executive Officer the IITRE does not adequately address an identified toxicity problem.
- 6) The Discharger shall assure that adequate resources are available to implement the required TRE/TIE.

3. Best Management Practices and Pollution Prevention

a. Pollutant Minimization Program

- 1) The Discharger shall develop and conduct a Pollutant Minimization Program (PMP) as further described below when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL, sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or aquatic organism tissue sampling) that a priority pollutant is present in the effluent above an effluent limitation and either:
 - a) A sample result is reported as DNQ and the effluent limitation is less than the RL; or
 - b) A sample result is reported as ND and the effluent limitation is less than the MDL.
- 2) The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:
 - a) An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
 - b) Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
 - c) Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
 - d) Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
 - e) An annual status report that shall be sent to the Regional Water Board including:
 - (1) All PMP monitoring results for the previous year;
 - (2) A list of potential sources of the reportable priority pollutant(s);
 - (3) A summary of all actions undertaken pursuant to the control strategy; and
 - (4) A description of actions to be taken in the following year.

4. Construction, Operation and Maintenance Specifications

- a. The Discharger's wastewater treatment plant shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to Title 23, Division 3, Chapter 14, California Code of Regulations.
- b. The Discharger shall provide safeguards to assure that should there be reduction, loss, or failure of electric power, the Discharger will comply with the requirements of this Order.

- c. The Discharger shall update as necessary, the "Operation and Maintenance Manual (O&M Manual)" which it has developed for the treatment facility to conform to latest plant changes and requirements. The O&M Manual shall be readily available to operating personnel onsite. The O&M Manual shall include the following:
- 1) Description of the treatment plant table of organization showing the number of employees, duties and qualifications and plant attendance schedules (daily, weekends and holidays, part-time, etc). The description should include documentation that the personnel are knowledgeable and qualified to operate the treatment facility so as to achieve the required level of treatment at all times.
 - 2) Detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation and equipment.
 - 3) Description of laboratory and quality assurance procedures.
 - 4) Process and equipment inspection and maintenance schedules.
 - 5) Description of safeguards to assure that, should there be reduction, loss, or failure of electric power, the Discharger will be able to comply with requirements of this Order.
 - 6) Description of preventive (fail-safe) and contingency (response and cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. These plans shall identify the possible sources (such as loading and storage areas, power outage, waste treatment unit failure, process equipment failure, tank and piping failure) of accidental discharges, untreated or partially treated waste bypass, and polluted drainage.

5. Special Provisions for Municipal Facilities (POTWs Only)

- a. The Discharger's collection system is part of the system that is subject to this Order. As such, the Discharger must properly operate and maintain its collection system (40 C.F.R. § 122.41(e)). The Discharger must report any non-compliance (40 C.F.R. § 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 C.F.R. § 122.41(d)). See the Order at Standard Provision VI.A.2.b. and Attachment D, subsections I.D, V.E, V.H, and I.C.

Furthermore, the General Waste Discharge Requirements for Collection System Agencies (Order No. 2006-0003 DWQ) contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. While the Discharger must comply with both Order No. 2006-0003 DWQ and this Order, the General Collection System WDR more clearly and specifically stipulates requirements for operation and maintenance and for reporting and mitigating sanitary sewer overflows. The Discharger and other governmental agencies that are discharging wastewater into the facility are required to obtain enrollment for regulation under Order No. 2006-0003-DWQ DWQ as soon as possible but no later than March 1, 2007.

b. Sludge Disposal Requirements

- 1) Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with State Water Board and Integrated Waste Management Board's joint regulations (Title 27) of the California Code of Regulations and approved by the Regional Water Board's Executive Officer.
- 2) The use and disposal of biosolids shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
- 3) Any proposed change in biosolids use or disposal practice from a previously approved practice should be reported to the Executive Officer and EPA Regional Administrator at least 90 days in advance of the change.
- 4) The Discharger shall take all reasonable steps to minimize or prevent any discharge or biosolids use or disposal that has the potential of adversely affecting human health or the environment.

c. The Discharger shall submit a monthly report that validates that tertiary treated effluent or recycled water for reuse is an oxidized, filtered, and disinfected wastewater. The report shall include:

- 1) When filtration through natural undisturbed soils or a bed of filter media, description of when, how often and whether coagulation of the wastewater is employed in the treatment process. If coagulation is not used at all times, the Discharger shall:
 - a) Continuously monitor the turbidity of the influent to the filters. Turbidity exceedances of 10 NTU or above at any time, and of 5 NTU for more than 15 minutes, shall be included in the monthly report;
 - b) Certify that chemical addition for coagulation has been automatically employed when the filter influent turbidity exceeds 5 NTU for more than 15 minutes.
- 2) Description of the type and rate of filtration employed in the treatment process.
- 3) Description of disinfection process utilized including chlorine or UV dosage as appropriate.

d. Pretreatment Program

- 1) The Discharger shall update as necessary and implement an acceptable pretreatment program.

- 2) The Discharger shall update as necessary the appropriate contractual agreements with all governmental agencies¹³. The contractual agreements shall give the Discharger the authority to implement and enforce the EPA approved pretreatment program within the sewer service areas of the treatment facility. The Discharger shall assure that any other steps necessary to provide this implementation and enforcement authority (e.g. adoption of ordinances, etc.) are taken by all governmental agencies. If a governmental agency has an EPA approved pretreatment program for any portion of the service area of the treatment facility, the Discharger's pretreatment program shall contain provisions ensuring that that governmental agency's program is implemented. In the event that any agency discharging to Discharger's facility fails to effectively implement its individual EPA approved pretreatment program, the Discharger shall implement and enforce its approved program within that agency's service area.
- 3) The Discharger shall ensure that the POTW¹⁴ pretreatment program for all contributory agencies discharging to the Discharger's treatment facility are implemented and enforced. The Discharger shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR 403, including any subsequent regulatory revisions to Part 403. Where Part 403 or subsequent revisions place mandatory actions upon the Discharger as Control Authority but does not specify a timetable for completion of the actions, the Discharger shall submit for approval of the Regional Water Board's Executive Officer, a schedule for implementation of the required actions and shall implement the approved schedule. The schedule for implementation shall be submitted within six months from the date that such mandatory actions are established. For violations of pretreatment requirements, the Discharger shall be subject to enforcement actions, penalties, fines and other remedies by the EPA, or other appropriate parties, as provided in the CWA, as amended (33 USC 1351 et seq.). The EPA or the Regional Water Board may also initiate enforcement action against an industrial user (IU) for non-compliance with applicable standards and requirements as provided in the CWA.
- 4) The Discharger shall perform the pretreatment functions as required in 40 CFR Part 403 including, but not limited to:
 - a) Enforce the pretreatment requirements under 40 CFR 403.5 and 403.6;
 - b) Implement the necessary legal authorities as provided in 40 CFR 403.8(f)(1);
 - c) Implement the programmatic functions as provided in 40 CFR 403.8(f)(2);

¹³ Member agencies and sewerage agencies discharging wastewater into the facility.

¹⁴ Publicly owned treatment works.

- d) Publish a list of significant non-compliance as required by 40 CFR 403.8(f)(2)(vii); and
 - e) Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR 403.8(f)(3).
- 5) The following wastes shall not be introduced into the treatment works:
- a) Wastes which create a fire or explosion hazard in the treatment works;
 - b) Wastes which will cause corrosive structural damage to treatment works, but, in no case, wastes with a pH lower than 5.0 unless the works are designed to accommodate such wastes;
 - c) Wastes at a flow rate and/or pollutant discharge rate which is excessive over relatively short time periods so that there is a treatment process upset and subsequent loss of treatment efficiency;
 - d) Solid or viscous wastes in amounts that would cause obstruction to the flow in sewers or otherwise interfere with the proper operation of the treatment works.
- 6) The Discharger shall ensure compliance with any existing or future pretreatment standard promulgated by EPA under Section 307 of the CWA or amendments thereto for any discharge to the municipal system.
- 7) The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement.
- 8) The Discharger shall require each user not in compliance with any pretreatment standard to submit periodic notice (over intervals not to exceed nine months) of progress toward compliance with applicable toxic and pretreatment standards developed pursuant to the CWA or amendments thereto. The Discharger shall forward a copy of such notice to the Regional Water Board and to the EPA Regional Administrator.

6. Other Special Provision - Not Applicable

7. Compliance Schedules – DP 001 and DP 002

- a. On December 20, 2006, the Discharger submitted a report that proposes a plan and schedule for achieving compliance with free cyanide, total recoverable mercury, and Bis(2-ethylhexyl) phthalate effluent limitations. The Discharger shall implement that plan and schedule upon approval by the Executive Officer. Quarterly progress reports are also required to be submitted.

- b. Violation(s) of interim effluent limitations are subject to the same enforcement remedies provided in the Water Code for violation(s) of final effluent limits.
- c. This Order will be reopened to consider appropriate changes to the compliance schedule if and as necessary based on submission of additional justification.

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in Section IV of this Order will be determined as specified below:

A. General.

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

B. Average Monthly Effluent Limitation (AMEL).

If the average (or when applicable, the median determined by multiple sample data, see Attachment E, paragraph X.B.5.) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation, though the Discharger may be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger may be considered out of compliance for that calendar month. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month.

C. Average Weekly Effluent Limitation (AWEL).

If the average (or when applicable, the median for multiple sample data, see Attachment E, paragraph X.B.5.) of daily discharges over a calendar week exceeds the AWEL for a given parameter, this will represent a single violation, though the Discharger may be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger may be considered out of compliance for that calendar week. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week.

D. Maximum Daily Effluent Limitation (MDEL).

If a daily discharge (or when applicable, the median for multiple sample data of a daily discharge, see Attachment E, paragraph X.B.5.) exceeds the MDEL for a given parameter, the Discharger will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no compliance determination can be made for that day.

E. Instantaneous Minimum Effluent Limitation.

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

F. Instantaneous Maximum Effluent Limitation.

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, the Discharger may be considered out of compliance for that parameter for that single sample. Non-compliance for each sample may be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

G. 12-Month Running Average Effluent Limitation (12-MRAEL).

Compliance with the 12-month running average limits under Discharge Specification IV.A.1.c., IV.A.1.d., IV.C.1.b., and IV.C.1. c. shall be determined by the arithmetic mean of the last twelve monthly averages.

H. Total Chlorine Residual Limitation (TCR).

Compliance determinations for total chlorine residual shall be based on 99% compliance. To determine 99% compliance with the effluent limitation for total chlorine residual, the following conditions shall be satisfied:

1. The total time during which the total chlorine residual values are above 0.1 mg/L (instantaneous maximum value) shall not exceed 7 hours and 26 minutes in any calendar month;
2. No individual excursion from 0.1 mg/L value shall exceed 30 minutes; and
3. No individual excursion shall exceed 5.0 mg/L.

I. Turbidity Limitations.

The Discharger shall be considered in compliance with Discharge Specifications IV.A.1.e. and IV.C.1.d. if the following conditions are met. If the Discharger is using a properly operating backup turbidimeter, the reading of the backup turbidimeter shall be considered in determining whether there has been an actual noncompliance:

1. There are no excursions above the limits specified in Discharge Specifications IV.A.1.e.(1)(a) and (b) and IV.C.1.d.(1)(a);
2. Exceedances of the "10 NTU at any time" turbidity requirement do not exceed a duration of one minute.
3. The apparent exceedance was caused by interference with, or malfunction of, the monitoring instrument.

J. Coliform Organism Effluent Limitations.

1. Compliance with the average weekly total coliform limit expressed in Discharge Specification IV.A.1.e.4), IV.C.1.d.2), and IV.C.1.e shall be based on a median of test results from the previous 7 days. To comply with the limit, the 7-day median MPN must not exceed 2.2 per 100 milliliters on any day during the week. However, only one violation is recorded for each calendar week, even if the 7-day median MPN value is greater than 2.2 for more than one day in the week.
2. Compliance with the average weekly total coliform limit expressed in Discharge Specification IV.C.1.f. shall be based on a median of test results from the previous 7 days. To comply with the limit, the 7-day median MPN must not exceed 23 per 100 milliliters on any day during the week. However, only one violation is recorded for each calendar week, even if the 7-day median MPN value is greater than 23 for more than one day in the week.

K. pH Effluent Limitations.

Pursuant to 40 CFR 401.17, the Discharger shall be in compliance with the pH limitations specified in the Discharge Specification IV.A.1.a, above, provided that both of the following conditions are satisfied:

1. The total time during which the pH values are outside the required range of 6.5-8.5 pH values shall not exceed 7 hours and 26 minutes in any calendar month; and
2. No individual excursion from the range of pH values shall exceed 60 minutes.

L. TDS Increment Limit.

Compliance with Discharge Specifications IV.A.1.c.2) shall be based on flow weighted TDS water supply quality and shall be determined from TDS analysis of influent or secondary treated wastewater. The Discharger shall provide the necessary calculations showing the overall TDS water supply quality.

M. Priority Pollutants.

The Discharger shall be deemed out of compliance with an effluent limitation if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation.

1. Compliance determination shall be based on the reporting level selected from minimum level (ML)¹⁵ specified in Attachment "H" of this Order, unless an alternative reporting level is approved by the Regional Water Board's Executive Officer. When there is more than one ML value for a given substance, the Discharger shall select the ML value that is below the calculated effluent limitation, and use its associated analytical method, listed in Attachment "H" of this Order. If no ML value is below the effluent limitation, then the Regional Water Board will select as the reporting level the lowest ML value and its associated analytical method.
2. When determining compliance with an average monthly limit and more than one sample result is available in a month, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of detected but not quantified (DNQ) or not detected (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ. If a sample result, or the arithmetic mean or median of multiple sample results, is below the reporting level, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the Discharger conducts a pollutant minimization program (PMP)¹⁶ the Discharger shall not be deemed out of compliance.

N. Non-Priority Pollutants.

The discharge shall be considered to be in compliance with an effluent limitation that is less than or equal to the method detection limit (MDL) specified in 40 CFR 136 if the arithmetic mean of all test results for the monitoring period is less than the constituent effluent limitation. Analytical results that are less than the specified MDL shall be assigned a value of zero.

¹⁵ Minimum level is the concentration at which the entire analytical system must give a recognizable signal and acceptable point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

¹⁶ The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation.

O. Compliance Determination

Compliance determinations shall be based on available analyses for the time interval associated with the effluent limitation. Where only one sample analysis is available in a specified time interval (e. g., monthly or weekly average), that sample shall serve to characterize the discharge for the entire interval. If quarterly sample results show noncompliance with the average monthly limit and that sample result is used for compliance determinations for each month of the quarter, then three separate violations of the average monthly limit shall be deemed to have occurred.

Compliance with a single effluent limitation which applies to a group of chemicals (e.g., PCBs), based on a single sample shall be determined by considering the concentrations of individual members of the group to be zero if the analytical response for the individual chemical falls below the method detection limit for that chemical.

ATTACHMENT A – DEFINITIONS

Arithmetic Mean (μ), also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n$$

where: Σx is the sum of the measured ambient water concentrations, and
 n is the number of samples.

Average Monthly Effluent Limitation (AMEL): the highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL): the highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Best Management Practices (BMPs) are methods, measures, or practices designed and selected to reduce or eliminate the discharge of pollutants to surface waters from point and nonpoint source discharges including storm water. BMPs include structural and non-structural controls, and operation and maintenance procedures, which can be applied before, during, and/or after pollution producing activities.

Bioaccumulative pollutants are those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Carcinogenic pollutants are substances that are known to cause cancer in living organisms.

Coefficient of Variation (CV) is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Criteria Continuous Concentration (CCC) equals the highest concentration of a pollutant to which aquatic life can be exposed for an extended period of time (4 days) without deleterious effects.

Criteria Maximum Concentration (CMC) equals the highest concentration of a pollutant to which aquatic life can be exposed for a short period of time without deleterious effects.

Daily Discharge: Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

Detected, but Not Quantified (DNQ) are those sample results less than the RL, but greater than or equal to the laboratory's MDL.

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Dilution Ratio is the critical low flow of the upstream receiving water divided by the flow of the effluent discharged.

Effluent Concentration Allowance (ECA) is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Estimated Chemical Concentration is the estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Existing Discharger means any discharger that is not a new discharger. An existing discharger includes an "increasing discharger" (i.e., an existing facility with treatment systems in place for its current discharge that is or will be expanding, upgrading, or modifying its existing permitted discharge after the effective date of this Policy).

Infeasible means not capable of being accomplished in a successful manner within a reasonable period of time, taking into account economic, environmental, legal, social, and technological factors.

Inland Surface Waters are all surface waters of the State that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation: the highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation: the lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Load Allocation (LA) is the portion of receiving water's total maximum daily load that is allocated to one of its nonpoint sources of pollution or to natural background sources.

Maximum Daily Flow is the maximum flow sample of all samples collected in a calendar day.

Maximum Daily Effluent Limitation (MDEL) means the highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

MEC: Maximum Effluent Concentration.

Median is the middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL) is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR 136, Appendix B, revised as of May 14, 1999.

Minimum Level (ML) is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

Not Detected (ND) are those sample results less than the laboratory's MDL.

Objectionable Bottom Deposits are an accumulation of materials or substances on or near the bottom of a water body, which creates conditions that adversely impact aquatic life, human health, beneficial uses, or aesthetics. These conditions include, but are not limited to, the accumulation of pollutants in the sediments and other conditions that result in harm to benthic organisms, production of food chain organisms, or fish egg development. The presence of such deposits shall be determined by RWQCB(s) on a case-by-case basis.

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP) means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to CWC Section 13263.3(d), shall be considered to fulfill the PMP requirements. The following reporting protocols and definitions are used in determining the need to conduct a Pollution Minimization Program (PMP). Reporting protocols in the Monitoring and Reporting Program, Attachment E, Section X.B.4 describe sample results that are to be reported as Detected but Not Quantified (DNQ) or Not Detected (ND). Definitions for a Minimum Level (ML) and Method Detection Limit (MDL) are provided in Attachment A. A Reporting Level (RL) is the ML associated with an analytical method selected by the Discharger that is authorized for monitoring effluent limitations under this Order.

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code Section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the SWRCB or RWQCB.

Process Optimization means minor changes to the existing facility and treatment plant operations that optimize the effectiveness of the existing treatment processes.

Public Entity includes the federal government or a state, county, city and county, city, district, public authority, or public agency.

Reporting Level (RL) is the ML corresponding to an approved analytical method for reporting a sample result that is selected either from Appendix 4 of the SIP by the Regional Water Board in accordance with Section 2.4.2 of the SIP or established in accordance with Section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Source of Drinking Water is any water designated as municipal or domestic supply (MUN) in a RWQCB basin plan.

Standard Deviation (σ) is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

μ is the arithmetic mean of the observed values; and

n is the number of samples.

Toxicity Reduction Evaluation (TRE) is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

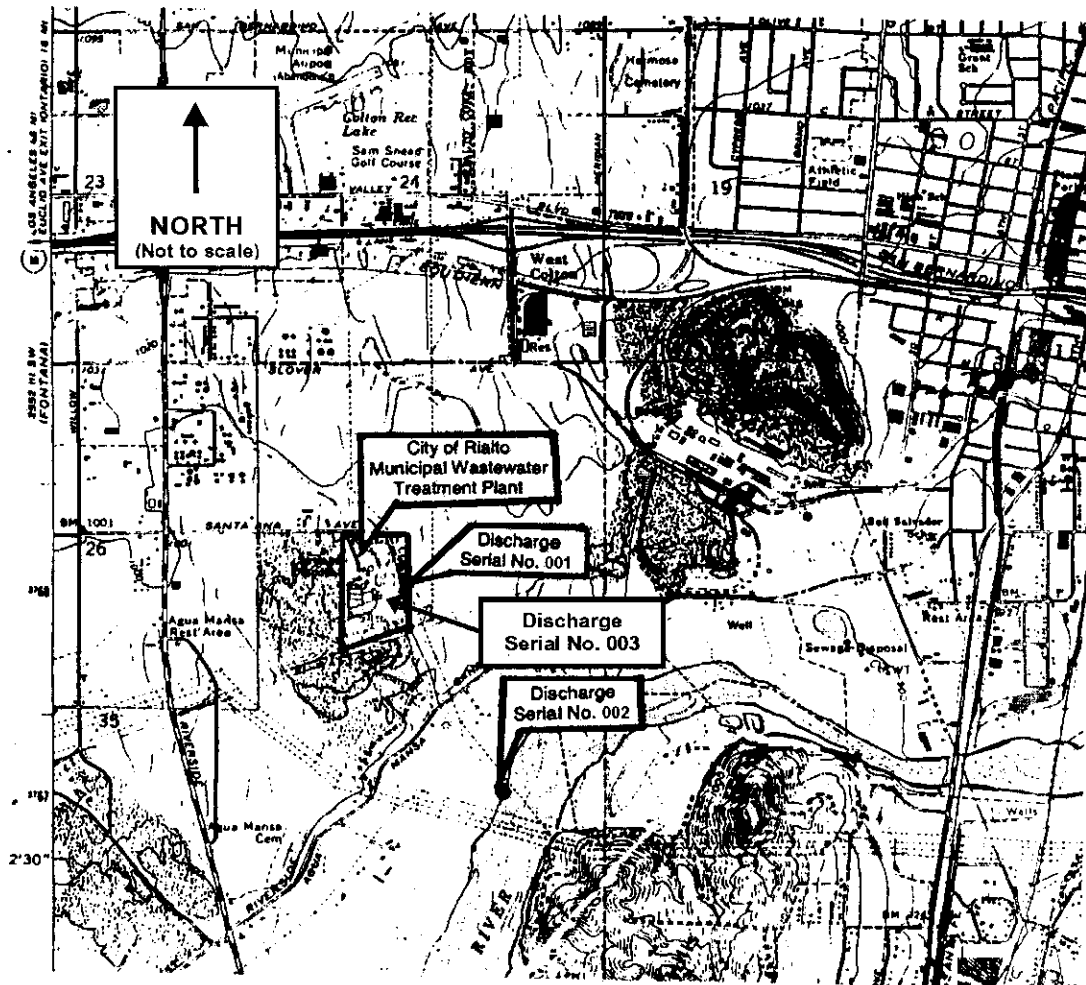
Use Attainability Analysis is a structured scientific assessment of the factors affecting the attainment of the use which may include physical, chemical, biological and economic factors as described in 40 CFR 131.10(g) (40 CFR 131.3, revised as of July 1, 1997).

Water Effect Ratio (WER) is an appropriate measure of the toxicity of a material obtained in a site water divided by the same measure of the toxicity of the same material obtained simultaneously in a laboratory dilution water.

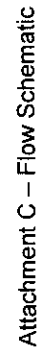
12-Month Running Average Effluent Limitation (12-MRAEL): the highest allowable average of monthly discharges over last twelve months, calculated as the sum of all monthly discharges measured during last twelve months divided by the number of monthly discharges measured during that time period.

ATTACHMENT B – LOCATION MAP

RIALTO WATER RECLAMATION FACILITY LOCATION MAP



RIAL TO WATER RECLAMATION FACILITY TREATMENT PROCESS FLOW DIAGRAM



ATTACHMENT D – STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code (CWC) and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification, or denial of a permit renewal application [40 CFR §122.41(a)].
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement [40 CFR §122.41(a)(1)].

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order [40 CFR §122.41(c)].

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment [40 CFR §122.41(d)].

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order [40 CFR §122.41(e)].

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges [40 CFR §122.41(g)].

2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of State or local law or regulations [40 CFR §122.5(c)].

F. Inspection and Entry

The Discharger shall allow the Regional Water Quality Control Board (RWQCB), State Water Resources Control Board (SWRCB), United States Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to [40 CFR §122.41(i)] [CWC 13383(c)]:

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order [40 CFR §122.41(i)(1)];
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order [40 CFR §122.41(i)(2)];
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order [40 CFR §122.41(i)(3)];
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the CWC, any substances or parameters at any location [40 CFR §122.41(i)(4)].

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility [40 CFR §122.41(m)(1)(i)].
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production [40 CFR §122.41(m)(1)(ii)].
2. Bypass not exceeding limitations – The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below [40 CFR §122.41(m)(2)].

3. Prohibition of bypass – Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless [40 CFR §122.41(m)(4)(i)]:
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage [40 CFR §122.41(m)(4)(A)];
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance [40 CFR §122.41(m)(4)(B)]; and
 - c. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below [40 CFR §122.41(m)(4)(C)].
4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above [40 CFR §122.41(m)(4)(ii)].
5. Notice
 - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass [40 CFR §122.41(m)(3)(i)].
 - b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice) [40 CFR §122.41(m)(3)(ii)].

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the permittee. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation [40 CFR §122.41(n)(1)].

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review [40 CFR §122.41(n)(2)].

2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that [40 CFR §122.41(n)(3)]:
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset [40 CFR §122.41(n)(3)(i)];
 - b. The permitted Facility was, at the time, being properly operated [40 CFR §122.41(n)(3)(i)];
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b. below (24-hour notice) [40 CFR §122.41(n)(3)(iii)]; and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above [40 CFR §122.41(n)(3)(iv)].
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof [40 CFR §122.41(n)(4)].

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition [40 CFR §122.41(f)].

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit [40 CFR §122.41(b)].

C. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the CWC [40 CFR §122.41(l)(3)] [40 CFR §122.61].

III. STANDARD PROVISIONS – MONITORING

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity [40 CFR §122.41(j)(1)].
- B.** Monitoring results must be conducted according to test procedures under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503 unless other test procedures have been specified in this Order [40 CFR §122.41(j)(4)] [40 CFR §122.44(i)(1)(iv)].

IV. STANDARD PROVISIONS – RECORDS

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 CFR Part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time [40 CFR §122.41(j)(2)].
- B. Records of monitoring information shall include:**
 - 1. The date, exact place, and time of sampling or measurements [40 CFR §122.41(j)(3)(i)];
 - 2. The individual(s) who performed the sampling or measurements [40 CFR §122.41(j)(3)(ii)];
 - 3. The date(s) analyses were performed [40 CFR §122.41(j)(3)(iii)];
 - 4. The individual(s) who performed the analyses [40 CFR §122.41(j)(3)(iv)];
 - 5. The analytical techniques or methods used [40 CFR §122.41(j)(3)(v)]; and
 - 6. The results of such analyses [40 CFR §122.41(j)(3)(vi)].
- C. Claims of confidentiality for the following information will be denied [40 CFR §122.7(b)]:**
 - 1. The name and address of any permit applicant or Discharger [40 CFR §122.7(b)(1)]; and

2. Permit applications and attachments, permits and effluent data [40 CFR §122.7(b)(2)].

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the Regional Water Board, SWRCB, or USEPA within a reasonable time, any information which the Regional Water Board, SWRCB, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, SWRCB, or USEPA copies of records required to be kept by this Order [40 CFR §122.41(h)] [CWC 13267].

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below [40 CFR Section 122.41(k)].
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA) [40 CFR Section 122.22(a)(3)].
3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above [40 CFR Section 122.22(b)(1)];
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) [40 CFR Section 122.22(b)(2)]; and
 - c. The written authorization is submitted to the Regional Water Board, and State Water Board [40 CFR Section 122.22(b)(3)].

4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board, State Water Board or USEPA prior to or together with any reports, information, or applications, to be signed by an authorized representative [40 CFR Section 122.22(c)].
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations" [40 CFR §122.22(d)].

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order [40 CFR §122.41(l)(4)].
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or SWRCB for reporting results of monitoring of sludge use or disposal practices [40 CFR §122.41(l)(4)(i)].
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board [40 CFR §122.41(l)(4)(ii)].
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order [40 CFR §122.41(l)(4)(iii)].

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date [40 CFR §122.41(l)(5)].

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance [40 CFR §122.41(l)(6)(i)].
2. The following shall be included as information that must be reported within 24 hours under this paragraph [40 CFR §122.41(l)(6)(ii)]:
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order [40 CFR §122.41(l)(6)(ii)(A)].
 - b. Any upset that exceeds any effluent limitation in this Order [40 CFR §122.41(l)(6)(ii)(B)].
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours [40 CFR §122.41(l)(6)(iii)].

F. Planned Changes

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when [40 CFR §122.41(l)(1)]:

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40 CFR §122.29(b) [40 CFR §122.41(l)(1)(i)]; or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants which are not subject to effluent limitations in this Order. [40 CFR Section 122.41(l)(1)(ii)].
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan [40 CFR §122.41(l)(1)(iii)].

G. Anticipated Noncompliance

The Discharger shall give advance notice to the Regional Water Board or SWRCB of any planned changes in the permitted facility or activity that may result in noncompliance with General Order requirements [40 CFR §122.41(l)(2)].

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above [40 CFR Section 122.41(l)(7)].

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, SWRCB, or USEPA, the Discharger shall promptly submit such facts or information [40 CFR §122.41(l)(8)].

VI. STANDARD PROVISIONS – ENFORCEMENT

- A. The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the CWC, including, but not limited to, sections 13385, 13386, and 13387.

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following [40 CFR Section 122.42(b)]:

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to Sections 301 or 306 of the CWA if it were directly discharging those pollutants [40 CFR Section 122.42(b)(1)]; and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order [40 CFR Section 122.42(b)(2)].
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW [40 CFR Section 122.42(b)(3)].

Attachment E – Monitoring and Reporting Program

Table of Contents

Attachment E – Monitoring and Reporting Program (MRP)	E-3
I. General Monitoring Provisions	E-3
A. General Monitoring Provision	E-3
II. Monitoring Locations	E-7
III. Influent Monitoring Requirements	E-8
A. Monitoring Location at M-INF	E-8
IV. Effluent Monitoring Requirements	E-10
A. Tertiary Effluent Monitoring Locations at M-001 and M-002	E-10
B. Secondary Effluent Monitoring Locations at M-001 & M-002 With 20:1 Dilution	E-13
V. Whole Effluent Toxicity Testing Requirements	E-13
A. Toxicity Monitoring Requirements at M-001 and M-002	E-13
VI. Land Discharge Monitoring Requirements – Not Applicable	E-15
VII. Reclamation Monitoring Requirements	E-16
A. Monitoring Location REC-001	E-16
B. Monitoring Users	E-16
VIII. Receiving Water Monitoring Requirements – Surface Water and Groundwater	E-17
A. Monitoring During 20:1 Dilution	E-17
B. Monitoring Location R-001U for Surface Water	E-17
C. Monitoring Location R-001D for Surface Water	E-17
D. Regional Monitoring for Fish Flesh Testing	E-18
E. Groundwater—Not Applicable	E-18
IX. Other Monitoring Requirements	E-18
A. Biosolids Monitoring at B-001	E-18
B. Stormwater Monitoring – Not Applicable	E-18
C. Water Supply Monitoring	E-19
D. Pretreatment Monitoring and Reporting	E-19
X. Reporting Requirements	E-22
A. General Monitoring and Reporting Requirements	E-22
B. Self Monitoring Reports (SMRs)	E-24
C. Discharge Monitoring Reports (DMRs)	E-27
D. Other Reports – Not Applicable	E-27

List of Tables

Table 1. Annual Sampling Schedule	E-7
Table 2. Monitoring Station Locations	E-8
Table 3. Influent Monitoring Requirements	E-8
Table 4. Tertiary Effluent Monitoring Requirements at M-001 & M-002	E-10
Table 5. Secondary Effluent Monitoring at M-001 & M-002 with 20:1 Dilution	E-13
Table 6. Reclamation Monitoring Requirements	E-16
Table 7. Receiving Water Monitoring at R-001U	E-17
Table 8. Receiving Water Monitoring at R-001D	E-17
Table 9. Biosolids Monitoring at B-001	E-18
Table 10. Reporting Requirements	E-23
Table 11. Monitoring and Reporting Schedule	E-24

ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

The Code of Federal Regulations (CFR) at 40 CFR §122.48 requires that all NPDES permits specify monitoring and reporting requirements. CWC Sections 13267 and 13383 also authorize the Regional Water Quality Control Board (RWQCB) to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement the federal and California regulations.

I. GENERAL MONITORING PROVISIONS

A. General Monitoring Provision

1. All sampling and sample preservation shall be in accordance with the current edition of *"Standard Methods for the Examination of Water and Wastewater"* (American Public Health Association).
2. All laboratory analyses shall be performed in accordance with test procedures under 40 CFR 136 (revised as of May 14, 1999) "Guidelines Establishing Test Procedures for the Analysis of Pollutants," promulgated by the United States Environmental Protection Agency (EPA), unless otherwise specified in this MRP. In addition, the Regional Water Board and/or EPA, at their discretion, may specify test methods that are more sensitive than those specified in 40 CFR 136. For priority pollutants, the test methods must meet the lowest minimum levels (MLs) specified in Attachment "H" of this Order, where no methods/MLs are specified in Attachment "H", then the monitoring is to be conducted in accordance with methods/MLs approved by the Regional Water Board or State Water Board consistent with the State Water Board's Quality Assurance Program.
3. Chemical, bacteriological, and bioassay analyses shall be conducted at a laboratory certified for such analyses by the California Department of Health Services or EPA or at laboratories approved by the Regional Water Board's Executive Officer.
4. Whenever the Discharger monitors any pollutant more frequently than is required by this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the discharge monitoring report specified by the Executive Officer.
5. In conformance with federal regulations 40 CFR 122.45(c), analyses to determine compliance with the effluent limitations for metals shall be conducted using the total recoverable method. For Chromium (VI), the dissolved method in conformance with 40 CFR 136 may be used to measure compliance with the Chromium (VI) limitation.

6. For effluent wastewater monitoring:

- a. The discharger shall require its testing laboratory to calibrate the analytical system down to the minimum level (ML)¹ specified in Attachment "H" for priority pollutants with effluent limitations in this Order, unless an alternative minimum level is approved by the Regional Water Board's Executive Officer. When there is more than one ML value for a given substance, the discharger shall use the ML values, and their associated analytical methods, listed in Attachment "H" that are below the calculated effluent limitation. The discharger may select any one of those cited analytical methods for compliance determination. If no ML value is below the effluent limitation, then the lowest ML value and its associated analytical method, listed in Attachment "H" shall be used. Any internal quality control data associated with the sample must be reported when requested by the Executive Officer. The Regional Water Board will reject the quantified laboratory data if quality control data is unavailable or unacceptable.
- b. The discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:
 - 1) Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
 - 2) Sample results less than the reported ML, but greater than or equal to the laboratory's current Method Detection Limit (MDL)², shall be reported as "Detected, but Not Quantified," or "DNQ." The estimated chemical concentration of the sample shall also be reported.
 - 3) Sample results not detected above the laboratory's MDL shall be reported as "not detected" or "ND."
- c. The Discharger shall submit to the Regional Water Board reports necessary to determine compliance with effluent limitations in this Order and shall follow the chemical nomenclature and sequential order of priority pollutant constituents shown in Attachment "G" – Priority Pollutant Lists. The Discharger shall report with each sample result:
 - 1) The reporting level achieved by the testing laboratory; and

¹ Minimum level is the concentration at which the entire analytical system must give a recognizable signal and acceptable point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

² MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analytical concentration is greater than zero, as defined in 40 CFR 136, Appendix B, revised as of May 14, 1999.

- 2) The laboratory's current MDL, as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999).
- d. For receiving water monitoring and for those priority pollutants without effluent limitations, the Discharger shall require its testing laboratory to quantify constituent concentrations to the lowest achievable MDL as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999). In situations where the most stringent applicable receiving water objective (freshwater or human health (consumption of organisms only), as specified for that pollutant in 40 CFR 131.38³ is below the minimum level value specified in Attachment "H" and the Discharger cannot achieve an MDL value for that pollutant below the ML value, the Discharger shall submit justification why a lower MDL value cannot be achieved. Justification shall be submitted together with monthly monitoring reports.
7. For non-priority pollutants monitoring, all analytical data shall be reported with method detection limits, as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999).
8. The discharger shall have, and implement an acceptable written quality assurance (QA) plan for laboratory analyses. Duplicate chemical analyses must be conducted on a minimum of ten percent (10%) of the samples, or at least one sample per month, whichever is greater. A similar frequency shall be maintained for analyzing spiked samples. When requested by the Regional Water Board or EPA, the discharger will participate in the NPDES discharge monitoring report QA performance study.
9. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, the actions undertaken or proposed that will bring the discharge into full compliance with requirements at the earliest time, and an estimate of the date when the discharger will be in compliance. The discharger shall notify the Regional Water Board by letter when compliance with the time schedule has been achieved.
10. The discharger shall assure that records of all monitoring information are maintained and accessible for a period of at least five years (this retention period supercedes the retention period specified in Section IV.A. of Attachment D) from the date of the sample, report, or application. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge or by the request of the Regional Water Board at any time. Records of monitoring information shall include:
 - a The information listed in Attachment D- IV Standard Provisions – Records, subparagraph B. of this Order;

³

See *Federal Register*/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations.

- b The laboratory which performed the analyses;
 - c The date(s) analyses were performed;
 - d The individual(s) who performed the analyses;
 - e The modification(s) to analytical techniques or methods used;
 - f All sampling and analytical results, including
 - 1) Units of measurement used;
 - 2) Minimum reporting level for the analysis (minimum level);
 - 3) Results less than the reporting level but above the method detection limit (MDL);
 - 4) Data qualifiers and a description of the qualifiers;
 - 5) Quality control test results (and a written copy of the laboratory quality assurance plan);
 - 6) Dilution factors, if used; and
 - 7) Sample matrix type.
 - g All monitoring equipment calibration and maintenance records;
 - h All original strip charts from continuous monitoring devices;
 - i All data used to complete the application for this Order; and,
 - j Copies of all reports required by this Order.
 - k Electronic data and information generated by the Supervisory Control And Data Acquisition (SCADA) System.\
11. The flow measurement system shall be calibrated at least once per year or more frequently, to ensure continued accuracy.
12. All monitoring instruments and devices used by the discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. In the event that continuous monitoring equipment is out of service for greater than a 24-hour period, the discharger shall obtain a representative grab sample each day the equipment is out of service. The discharger shall correct the cause(s) of failure of the continuous monitoring equipment as soon as practicable. In its monitoring report, the discharger shall specify the period(s) during which the equipment was out of service and if the problem has not been corrected, shall identify the steps which the discharger is taking or proposes to take to bring the equipment back into service and the schedule for these actions.
13. Monitoring and reporting shall be in accordance with the following:
- a. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity.
 - b. The monitoring and reporting of influent, effluent, and sludge shall be done more frequently as necessary to maintain compliance with this Order and or as specified in this order.

- c. Whenever the discharger monitors any pollutant more frequently than is required by this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the discharge monitoring report specified by the Executive Officer.
- d. A "grab" sample is defined as any individual sample collected in less than 15 minutes.
- e. A composite sample is defined as a combination of no fewer than eight individual grab samples obtained over the specified sampling period. The volume of each individual grab sample shall be proportional to the discharge flow rate at the time of sampling. The compositing period shall equal the specific sampling period, or 24 hours, if no period is specified.
- f. Daily samples shall be collected on each day of the week.
- g. Monthly samples shall be collected on any representative day of each month.
- h. Quarterly samples: A representative sample shall be taken on any representative day of January, April, July, and October and test results shall be reported in either micrograms/liter (ug/L) or milligrams/liter (mg/L), as appropriate, by the last day of the month following the month that the sample was taken.
- i. Semi-annual samples shall be collected in January and July.
- j. Annual samples shall be collected in accordance with the following schedule:

Table 1. Annual Sampling Schedule

Year	Annual Samples
2007	January
2008	April
2009	July
2010	October
2011	January
2012	April

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table 2. Monitoring Station Locations

Discharge Point	Monitoring Location	Monitoring Location Description	Latitude and Longitude
--	M-INF	Influent diversion box.	34° 03' 18" N, 117° 21' 42" W
001	M-001	Final effluent downstream of dechlorination	34° 03' 04" N, 117° 21' 11" W
001	M-001A	Immediately downstream of filters	34° 03' 16" N, 117° 21' 37" W
001	M-001B	Discharge weir of Chlorine Contact Tank	34° 03' 12" N, 117° 21' 36" W
002	M-002	Emergency only.	34° 02' 36" N, 117° 21' 30" W
003	REC-001	Chlorine residual sampling point downstream of chlorine contact basin to recycled water pump station.	34° 03' 12" N, 117° 21' 36" W
--	R-001U	Receiving surface water – Santa Ana River at suitable location within 500 feet upstream of the outfall of DP 001	34° 02' 42" N, 117° 21' 08" W
--	R-001D	Receiving surface water – Santa Ana River within 500 feet downstream of the outfall of DP 001	34° 02' 32" N, 117° 21' 12" W

III. INFLUENT MONITORING REQUIREMENTS

A. Monitoring Location at M-INF

1. Sampling stations shall be established for the points of inflow to the treatment plant. The sampling station(s) shall be located upstream of any in-plant return flows and where representative sample(s) of the influent of the treatment plant can be obtained.
2. The Discharger shall monitor the influent to the facility at Monitoring Location M-INF as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level:

Table 3. Influent Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	See Section I.A.3, above, of this MRP
pH	pH Units	Recorder	Continuous	"
Specific Conductance	µmhos/cm	Recorder	Continuous	"
TOC	mg/L	Composite	Weekly	"
BOD ₅	mg/L	Composite	Weekly	"
Total Suspended Solids	mg/L	Composite	Weekly	"
Total Dissolved Solids	mg/L	Composite	Monthly	"
Ammonia-Nitrogen	mg/L	Grab	Monthly	"
Total Kjeldahl Nitrogen	mg/L	Composite	Monthly	"
Boron	mg/L	Composite	Quarterly	See Section I.A.3, above, of this MRP

Table 3. Influent Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Chloride	mg/L	Composite	Quarterly	"
Fluoride	mg/L	Composite	Quarterly	"
Sulfate	mg/L	Composite	Quarterly	"
Total Hardness	mg/L	Composite	Quarterly	"
Arsenic	µg/L	Composite	Quarterly	See Section I.A.2. above, of this MRP
Cadmium	µg/L	Composite	Quarterly	"
Total Chromium or Chromium VI	µg/L	Composite	Quarterly	See Section I.A.2. above, of this MRP
Copper	µg/L	Composite	Quarterly	"
Lead	µg/L	Composite	Quarterly	"
Mercury	µg/L	Composite	Quarterly	"
Nickel	µg/L	Composite	Quarterly	"
Selenium	µg/L	Composite	Quarterly	"
Silver	µg/L	Composite	Quarterly	"
Zinc	µg/L	Composite	Quarterly	"
Cyanide (Free)	µg/L	Grab	Quarterly	"
Bis(2-ethylhexyl) phthalate	µg/L	Grab	Quarterly	"
Volatile organic portion of EPA Priority Pollutants ⁴ (See Attachment G)	µg/L	Grab	Annually	See Section I.A.2. above, of this MRP
Remaining EPA Priority Pollutants ⁵ (See Attachment G)	µg/L	Composite	Annually	"

⁴ EPA priority pollutants are those remaining volatile organic pollutants listed in Attachment "G" which are not specifically listed in this monitoring program table.

⁵ Remaining EPA priority pollutants are those pollutants listed in Attachment "G" which are not volatile organics and pollutants not specifically listed in this monitoring program table.

IV. EFFLUENT MONITORING REQUIREMENTS

The Discharger shall monitor tertiary effluent at monitoring locations as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level.

A. Tertiary Effluent Monitoring Locations at M-001 and M-002

1. The Discharger shall monitor tertiary treated effluent for DP 001 and DP 002 at Monitoring Locations M-001 and M-002 as follows.

Table 4. Tertiary Effluent Monitoring Requirements at M-001 & M-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Test Method And Reporting Level
Flow	mgd	Recorder/ Totalizer	Continuous	See Section I.A.3. above, of this MRP
Specific Conductance	µmhos/cm	Recorder	Continuous	"
pH	pH units	"	Continuous	"
Total Residual Chlorine	mg/L	"	Continuous	"
Turbidity ⁶	NTU	Recorder	Continuous	"
CT ⁷	mg/L-min	Recorder	Continuous	See Section I.A.3., above, of this MRP
Coliform Organisms ⁸	MPN per 100 ml ⁹	Grab	Daily	See Section I.A.3., above of this MRP
Total Organic Carbon (TOC)	mg/L	Composite	Weekly	"
Ammonia-Nitrogen	mg/L	Grab	Twice-Weekly	See Section I.A.3. above, of this MRP
Temperature	°C	Grab	Weekly	"
BOD ₅	mg/L	Composite	Weekly	"
Suspended Solids	mg/L	Composite	Weekly	"
Total Dissolved Solids	"	"	Monthly	"
Total Hardness	"	Composite	Monthly	"
Total Inorganic Nitrogen	mg/L	Composite	Monthly	"
Nitrate Nitrogen	"	"	Monthly	"

⁶ Turbidity analysis shall be continuous, performed by a continuous recording turbidimeter. Compliance with the daily average operating filter effluent turbidity shall be determined by averaging the levels of recorded turbidity taken at a minimum of four-hour intervals over a 24-hour period. The results of the daily average turbidity determinations shall be reported monthly. Turbidity samples shall be collected at M-001A.

⁷ CT is the product of total chlorine residual and modal contact time measured at the same point.

⁸ Coliform samples shall be collected at M-001B.

⁹ MPN/100mL = Most Probable Number per 100 milliliters.

Table 4. Tertiary Effluent Monitoring Requirements at M-001 & M-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Test Method And Reporting Level
Toxicity Monitoring	TUc	See Section V, Below	Monthly	See Section V, Below
Copper	µg/L	Composite	Monthly	See Sections I.A.2., I.A.3., above of this MRP and RL 5 µg/L
Cyanide, free	µg/L	Grab	Monthly	See Sections I.A.2., I.A.3., above of this MRP and RL 5 µg/L
Mercury	µg/L	Composite	Monthly	See Sections I.A.2., I.A.3., above of this MRP and RL 0.05 µg/L
Bis (2-ethylhexyl) phthalate	µg/L	Grab	Monthly	See Sections I.A.2., I.A.3., above of this MRP and RL 5 µg/L
Bicarbonate	mg/L	Composite	Quarterly	See Section I.A.3. above, of this MRP
Boron	mg/L	Composite	Quarterly	"
Calcium	mg/L	Composite	Quarterly	"
Carbonate	mg/L	Composite	Quarterly	"
Chloride	mg/L	Composite	Quarterly	"
Fluoride	mg/L	Composite	Quarterly	See Section I.A.3. above, of this MRP
Magnesium	mg/L	Composite	Quarterly	"
Nitrate	mg/L	Composite	Quarterly	"
Sodium	mg/L	Composite	Quarterly	"
Sulfate	mg/L	Composite	Quarterly	"
Aluminum	mg/L	Composite	Quarterly	"
Antimony	mg/L	Composite	Quarterly	See Sections I.A.2., I.A.3., above of this MRP
Arsenic	µg/L	Composite	Quarterly, (See IV. A.2., below)	"
Barium	µg/L	Composite	Quarterly, (See IV. A.2., below)	"
Cadmium	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 0.5 µg/L
Chromium (VI) or Total Chromium ¹⁰	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3. above of this MRP and RL 5 µg/L, Total Cr, RL 2 µg/L
Cobalt	µg/L	Composite	Quarterly (See IV. A.2., below)	"

¹⁰ If Total Chromium test result is greater than 11 µg/L, the following sample shall be tested for Chromium VI, until directed otherwise.

Table 4. Tertiary Effluent Monitoring Requirements at M-001 & M-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Test Method And Reporting Level
Lead	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 5 µg/L
Nickel	µg/L	Composite	Quarterly (See IV. A.2., below)	"
Selenium	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 2 µg/L
Silver	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 1 µg/L
Zinc	µg/L	Composite	Quarterly (See IV. A.2., below)	See Sections I.A.2., I.A.3., above of this MRP
NDMA	µg/L	Composite	Annually (See IV. A.3., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 5 µg/L
Hexachlorocyclohexane-gamma	µg/L	Composite	Annually (See IV. A.3., below)	See Sections I.A.2., I.A.3., above of this MRP and RL 0.02 µg/L
Volatile organic portion of remaining EPA Priority Pollutants ⁶ (See Attachment "G")	µg/L	Grab	Annually (See IV. A.3., below)	See Sections I.A.2., I.A.3., above of this MRP
Remaining EPA Priority Pollutants ⁷ (See Attachment "G")	µg/L	Composite	Annually (See IV. A.3., below)	See Sections I.A.2., I.A.3., above of this MRP

2. The monitoring frequency for those priority pollutants that are detected during the required quarterly monitoring at a concentration greater than the concentration specified for that pollutant¹¹ in Attachment I shall be accelerated to monthly. To return to the monitoring frequency specified, the Discharger shall request and receive approval from the Regional Water Board's Executive Officer or designee.
3. The monitoring frequency for those priority pollutants that are detected during the required annual monitoring at a concentration greater than the concentration specified for that pollutant⁹ in Attachment I shall be accelerated to quarterly for one year. To return to the monitoring frequency specified, the Discharger shall request and receive approval from the Regional Water Board's Executive Officer or designee.

¹¹ For those priority pollutants without specified criteria values, accelerated monitoring is not required.

B. Secondary Effluent Monitoring Locations at M-001 & M-002 With 20:1 Dilution

1. The Discharger shall monitor secondary treated effluent at M-001 and M-002 when 20:1 dilution is provided by the Santa Ana River at the time of the discharge, as follows:

Table 5. Secondary Effluent Monitoring at M-001 & M-002 with 20:1 Dilution

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Test Method
Flow	mgd	Recorder/Totalizer	Continuous	See Section I.A.3., above, of this MRP
pH	pH units	Recorder/Totalizer	Continuous	See Section I.A.3., above, of this MRP
Total Residual Chlorine	mg/L	Recorder/Totalizer	Continuous	See Section I.A.3., above, of this MRP
BOD ₅	mg/L	Grab	Daily (when discharging)	"
Coliform Organisms	MPN per 100 ml ¹²	Grab	Daily (when discharging)	See Sections I.A.2., I.A.3. & I.B., above of this MRP
Suspended Solids	mg/L	Grab	Daily (when discharging)	See Sections I.A.2., I.A.3. & I.B., above of this MRP
EPA Priority Pollutants	µg/L	Grab	Annually ¹³ (See IV.A.3., above)	See Sections I.A.2., I.A.3. & I.B., above of this MRP

V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

A. Toxicity Monitoring Requirements at M-001 and M-002

1. The Discharger shall conduct critical life stage chronic toxicity testing in accordance with Method 1002.0 - Survival and Reproduction test for water flea, *Ceriodaphnia dubia* as specified in "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms", Fourth Edition, Environmental Monitoring Systems Laboratory, U.S. Environmental Protection Agency 2002, Cincinnati, Ohio (October 2002, EPA-821-R-02-013).
2. The Discharger shall establish procedures to ensure that the toxicity testing laboratory notifies the Discharger of the results of toxicity testing by the end of the next business day following the completion of such tests.

¹² MPN/100mL = Most Probable Number per 100 milliliters

¹³ Sample is collected from the first discharge, once a year.

3. A minimum of one monthly chronic toxicity test shall be conducted on representative composite samples.
4. The Discharger shall increase the frequency of chronic toxicity testing to every two weeks whenever any test result exceeds 1.0 TUc. The first test under the accelerated schedule shall be conducted within two weeks of receiving notice of the test that exceeds 1.0 TUc, and every two weeks thereafter. The Discharger may resume the regular test schedule when two consecutive chronic toxicity tests result in 1.0 TUc, or when the results of the Initial Investigation Reduction Evaluation conducted by the Discharger have adequately addressed the identified toxicity problem.
5. The presence of chronic toxicity shall be estimated as specified in Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. Fourth Edition. EPA-821-R-02-013.
6. Results for both survival and reproduction endpoints shall be reported in TUc, where $TUc = 100/NOEC$ or $100/IC_p$ or EC_p (p is the percent effluent). The no observed effect concentration (NOEC) is the highest concentration of toxicant to which organisms are exposed in a chronic test, that causes no observable adverse effect on the tests organisms (e.g., the highest concentration of toxicant to which the values for the observed responses are not statistically significant different from the controls). The inhibition concentration (IC) is a point estimate of the toxicant concentration that causes a given percent reduction in a non-quantal biological measurement (e.g., reproduction or growth) calculated from a continuous model (the EPA Interpolation Method). The effective concentration (EC) is a point estimate of the toxicant concentration that would cause a given percent reduction in quantal biological measurement (e.g., larval development, survival) calculated from a continuous model (e.g., probit).
7. Additional Testing Requirements
 - a. A series of at least five dilutions and a control will be tested. Five dilutions of the series shall be within 60% to 100% effluent concentration.
 - b. If organisms are not cultured in-house, concurrent testing with reference toxicants shall be conducted. Where organisms are cultured in-house, monthly reference toxicant testing is sufficient. Reference toxicants shall also be conducted using the same test conditions as the effluent toxicity test (e.g., same test duration, etc).
 - c. If either of the reference toxicant test or the effluent tests do not meet all test acceptability criteria as specified in the manual¹⁴, then the Discharger must re-sample and re-test within 14 days or as soon as the Discharger receives notification of failed tests.

¹⁴ Refers to USEPA Manual "Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. - 4th Ed., October 2002, EPA-821-R-02-013.

- d. Control and dilution water should be receiving water or lab water, as appropriate, as described in the manual. If the dilution water used is different from the culture water, a second control, using culture water shall also be used.
8. Quality Assurance/Control:
- a. A quality assurance/quality control (QA/QC) program shall be instituted to verify the results of the effluent toxicity monitoring program. The QA/QC program shall include but shall not be limited to the following: (1) Selection of an independent testing laboratory; (2) Approval by the Regional Board's Executive Officer or Executive Officer's designee of the independent testing laboratory; (3) Once during the year, the Discharger shall split samples with the independent laboratory for conducting chronic toxicity testing; (4) Results from the independent laboratory shall be submitted to the Regional Board and the Discharger for evaluation; (5) The Discharger shall review the test acceptability criteria in accordance with the EPA test protocols, Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. Fourth Edition. EPA-821-R-02-013.
 - b. Results from the independent laboratory of the annual QA/QC split samples are to be used for Quality Assurance/Quality Control (QA/QC) purposes only and not for purposes of determining compliance with other requirements of this Order.
9. The use of alternative methods for measuring chronic toxicity may be considered by the Executive Officer on a case-by-case basis. The use of a different test species, in lieu of conducting the required test species may be considered and approved by the Executive Officer on a case-by case basis upon submittal of the documentation supporting Discharger's determination that a different species is more sensitive and appropriate.
10. Reporting: Results of all toxicity testing conducted within the month following the reporting period shall be submitted monthly in accordance with "Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. Fourth Edition. EPA-821-R-02-013." The report shall include a determination of the median value of all chronic toxicity testing results conducted during the two previous months.
11. Whenever an Initial Investigation Reduction Evaluation is conducted, the results of the evaluation shall be submitted upon completion. In addition, monthly status reports shall be submitted as part of the Discharger's monitoring report for the previous month.

VI. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE.

VII. RECLAMATION MONITORING REQUIREMENTS

A. Monitoring Location REC-001

The Discharger shall monitor recycled wastewater at REC-001:

Table 6. Reclamation Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	---
pH	Standard units	Recorder/Totalizer	Continuous	--
CT ¹⁵	mg/L-min	Recorder	Continuous	See Section I.A.3., above, of this MRP
Turbidity ¹⁶	NTU	Recorder	Continuous	"
Coliform Organisms	MPN per 100 mL	Grab	Daily	"
BOD ₅	mg/L	Composite	Weekly	See Section I.A.3., above, of this MRP
Total Suspended Solids	mg/L	Composite	Weekly	See Section I.A.3., above, of this MRP
Total Inorganic Nitrogen	mg/L	Composite	Monthly	See Section I.A.3., above, of this MRP
TDS	mg/L	Composite	Monthly	See Section I.A.3., above, of this MRP

B. Monitoring Users

Whenever recycled water is supplied to a user, the Discharger shall record on a permanent log: the volume of recycled water supplied; the user of recycled water; the locations of those sites including the names of the groundwater management zones underlying the recycled water use sites; type of use (e.g. irrigation, industrial, etc); and the dates at which water is supplied. The Discharger shall submit annually a summary report of the recorded information by groundwater management zone to the Regional Water Board.

¹⁵ CT is the product of total chlorine residual and modal contact time measured at the same point.

¹⁶ Turbidity samples shall be collected at M-001A.

VIII. RECEIVING WATER MONITORING REQUIREMENTS – SURFACE WATER AND GROUNDWATER

A. Monitoring During 20:1 Dilution:

The Discharger shall make provisions for the measurement of the receiving water flow at a suitable location in the river and determine whether a 20:1 dilution exists at the point of discharge before discharging secondary treated effluent. A dilution of 20:1 or more exclusive of discharges to surface waters from upstream publicly owned treatment works is required at the point of discharge for discharge of secondary effluent. Flow measurements shall be made prior to any direct discharge to the river and shall continue on a daily basis until the discharge is terminated.

B. Monitoring Location R-001U for Surface Water

The Discharger shall monitor receiving water at R-001U as follows:

Table 7. Receiving Water Monitoring at R-001U

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Dissolved Oxygen	mg/L	Grab	Weekly	See Section I.A.3., above, of this MRP
Temperature	°C	"	Weekly	See Section I.A.3., above, of this MRP
pH	pH unit	Grab	Weekly	"
Hardness	mg/L	Grab	Monthly	See Section I.A.3., above, of this MRP
EPA Priority Pollutants	µg/L	"	Annually	See Section I.A.2., above, of this MRP

C. Monitoring Location R-001D for Surface Water:

The Discharger shall monitor receiving water at R-001D as follows:

Table 8. Receiving Water Monitoring at R-001D

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Dissolved Oxygen	mg/L	Grab	Weekly	See Section I.A.3., above, of this MRP
Temperature	°C	"	Weekly	See Section I.A.3., above, of this MRP

Table 8. Receiving Water Monitoring at R-001D

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
pH	pH unit	Grab	Weekly	"
Color change, foam, deposition of material, odor	--	Observe	Weekly	See Section I.A.3., above, of this MRP

D. Regional Monitoring for Fish Flesh Testing:

Unless otherwise directed by the Regional Water Board Executive Officer, the Discharger shall implement the approved plan for the annual sampling and testing of mercury levels in fish flesh samples collected from the Santa Ana River. The frequency of monitoring and submission of reports shall be as stipulated in the approved plan.

E. Groundwater—Not Applicable

IX. OTHER MONITORING REQUIREMENTS

A. Biosolids Monitoring at B-001

1. Biosolids monitoring shall be conducted as follows:

Table 9. Biosolids Monitoring at B-001

Biosolids Monitoring	Units	Type of Sample	Minimum Frequency of Sampling & Testing
Priority Pollutants	mg/kg	Grab	Semi-annually
Moisture Content (% solid)	mg/kg	Grab	Quarterly

2. The Discharger shall maintain a permanent log of solids hauled away from the treatment facilities for use/disposal elsewhere, including the date hauled, the volume or weight (in dry tons), type (screening, grit, raw sludge, biosolids), application (agricultural, composting, etc), and destination. This information shall be reported quarterly.

B. Stormwater Monitoring – Not Applicable

C. Water Supply Monitoring

1. In June of each year, a sample of each source of the water supplied to the sewered area shall be obtained and analyzed for total dissolved solids concentration expressed in "mg/L".
2. Monthly reports shall be submitted stating the amount (in percentage or acre-feet) supplied to the sewered area from each source of water and the resulting flow-weighted water supply quality for total dissolved solids.

D. Pretreatment Monitoring and Reporting

1. The Discharger shall submit to the Regional Water Board and the EPA Region 9, a quarterly compliance status report. The quarterly compliance status reports shall cover the periods January 1 - March 31, April 1 - June 30, July 1 - September 30, and October 1 - December 31. Each report shall be submitted by the end of the month following the quarter, except that the report for October 1 - December 31 may be included in the annual report. This quarterly reporting requirement shall commence for the first full quarter following issuance of this Order. The reports shall identify:
 - a. All significant industrial users (SIUs) which violated any standards or reporting requirements during that quarter;
 - b. The violations committed (distinguish between categorical and local limits);
 - c. The enforcement actions undertaken; and
 - d. The status of active enforcement actions from previous periods, including closeouts (facilities under previous enforcement actions which attained compliance during the quarter).
2. Annually, the Discharger shall submit a report to the Regional Water Board, the State Water Resources Control Board and the EPA Region 9 describing the pretreatment activities within the service area during the previous year. In the event that any control authority within the service area is not in compliance with any conditions or requirements of this Order or their approved pretreatment program (such as due to industrial user discharges, interjurisdictional agency agreement implementation issues, or other causes,) then the Discharger shall also include the reasons for non-compliance and state how and when the Discharger and the control authority shall comply with such conditions and requirements. This annual report shall cover operations from July 1 through June 30 of each fiscal year and is due on September 30 of each year. The report shall contain, but not be limited to, the following information:

- a. A summary of analytical results from representative, flow-proportioned, 24-hour composite sampling of the POTW's influent and effluent wastewaters for those pollutants which are known or suspected to be discharged by industrial users (IUs) as identified by EPA under Section 307(a) of the CWA. The summary will include the result of annual full priority pollutant scan, with quarterly samples analyzed only for those pollutants¹⁷ detected in the full scan. The Discharger shall also provide any influent or effluent monitoring data for non-priority pollutants which the Discharger believes may be causing or contributing to Interference, Pass Through or adversely impacting sludge quality. Sampling and analysis shall be performed in accordance with the techniques prescribed in 40 CFR 136 and amendments thereto.
- b. A discussion of any upset, interference, or pass-through incidents at the treatment plant (if any), which the Discharger knows or suspects were caused by IUs of the POTW system. The discussion shall include the following:
 - 1) The reasons why the incidents occurred, the corrective actions taken, and, if known, the name and address of the IU(s) responsible.
 - 2) A review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass through, interference or noncompliance with sludge disposal requirements.
- c. A complete and updated list of the Discharger's significant industrial users (SIUs), including names, Standard Industrial Classification (SIC) code(s) and addresses, and a list of any SIU deletions and/or additions. The Discharger shall provide a brief explanation for each deletion. The SIU list shall identify the SIUs subject to Federal Categorical Standards by specifying which set(s) of standards are applicable to each SIU. The list shall also indicate which SIUs are subject to local limitations more stringent than Federal Categorical Standards and those, which are not subject to local limits.
- d. A list or table characterizing the industrial compliance status of each SIU, including:
 - 1) SIU name;
 - 2) Industrial category;
 - 3) The type (processes) of wastewater treatment in place;
 - 4) Number of samples taken by the POTW during the year;
 - 5) Number of samples taken by the SIU during the year;
 - 6) Whether all needed certifications (if allowed) were provided by SIUs which have limits for total toxic organics;
 - 7) Federal and Regional Standards violated during the year, reported separately;

¹⁷ The Discharger is not required to analyze for asbestos.

- 8) Whether the SIU at any time in the year was in Significant Noncompliance (SNC)¹⁸, as defined by 40 CFR 403.12 (f)(2)(vii); and
 - 9) A summary of enforcement actions against the SIU taken during the year, including the type of action, final compliance date, and amount of fines assessed/collected (if any). Proposed actions, if known, should be included.
 - 10) Number of inspections conducted at each SIU during the year.
- e. A compliance summary table which includes:
- 1) SIU's which were in SNC at any time during the year;
 - 2) The total number of SIUs which are in SNC with pretreatment compliance schedules during the year;
 - 3) The total number of notices of violation and administrative orders issued against SIUs during the year;
 - 4) The total number of civil and criminal judicial actions filed against SIUs during the year;
 - 5) The number of SIUs which were published as being in SNC during the year; and
 - 6) The number of IUs from which penalties were collected during the year.
- f. A short description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to changes concerning:
- 1) The program's administrative structure;
 - 2) Local industrial discharge limitations;
 - 3) Monitoring program or monitoring frequencies;
 - 4) Legal authority or enforcement policy;
 - 5) Funding mechanisms; and
 - 6) Resource requirements and/or staffing levels.
- g. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases.
- h. A summary of public participation activities to involve and inform the public.
- i. A description of any changes in sludge disposal methods and a discussion of any concerns not described elsewhere in the report.
3. The cumulative number of industrial users that the Discharger has notified regarding Baseline Monitoring Reports and the cumulative number of industrial user responses.
 4. The Discharger shall submit the quarterly compliance status reports and the annual pretreatment report to EPA Region 9, the State Board and the Regional Water Board.

¹⁸ SNC is determined at the beginning of each quarter based on data of the previous six months.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. All analytical data shall be reported with method detection limit¹⁹ (MDLs) and with identification of either reporting level or limits of quantitation (LOQs).
3. Any internal quality control data associated with the sample must be reported when requested by the Executive Officer. The Regional Water Board will reject the quantified laboratory data if quality control data is unavailable or unacceptable.
4. Discharge monitoring data shall be submitted in a format acceptable by the Regional Water Board. Specific reporting format may include preprinted forms and/or electronic media. The results of all monitoring required by this Order shall be reported to the Regional Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this order.
5. The Discharger shall tabulate the monitoring data to clearly illustrate compliance and/or noncompliance with the requirements of the Order.
6. The Discharger shall submit to the Regional Water Board reports necessary to determine compliance with effluent limitations in this Order and shall follow the chemical nomenclature and sequential order of priority pollutant constituents shown in Attachment "G" – Priority Pollutant Lists. The Discharger shall report with each sample result:
 - a. The reporting level achieved by the testing laboratory; and
 - b. The laboratory's current MDL, as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999).
 - c. For receiving water monitoring and for those priority pollutants without effluent limitations, the Discharger shall require its testing laboratory to quantify constituent concentrations to the lowest achievable MDL as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999). In situations where the most stringent applicable receiving water objective (freshwater or human health (consumption of organisms only), as specified for that pollutant in 40 CFR 131.38²⁰ is below the minimum level value specified in Attachment "H" and the Discharger cannot achieve an MDL value for that pollutant below or equal to the ML value, the Discharger shall submit justification why a lower MDL value cannot be achieved. Justification shall be submitted together with monthly monitoring reports.

¹⁹ The standardized test procedure to be used to determine the method detection limit (MDL) is given at Appendix B, 'Definition and Procedure for the Determination of the Method Detection Limit' of 40 CFR 136.

²⁰ See *Federal Register*/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations.

7. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, and of the actions undertaken or proposed which will bring the discharge into full compliance with requirements at the earliest time, and an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Regional Water Board by letter when compliance with the time schedule has been achieved.
8. The reports for June and December shall include a roster of plant personnel, including job titles, duties, and level of State certification for each individual.
9. At any time during the term of this Order when electronic submittal of monitoring reports has become the norm, the State or Regional Water Board may notify the Discharger to discontinue submittal of hard copies of reports. When such notification is given, the Discharger shall stop submitting hard copies of required monitoring reports.
10. The Discharger shall report monitoring results for specific parameters in accordance with the following table:

Table 10. Reporting Requirements

Parameter	Measurement
Flow	Daily total flow
pH	Daily High and daily low
Total Residual Chlorine	Daily Maximum
Electrical Conductivity	Daily High
Turbidity	Daily maximum

11. The Discharger shall file a written report with the Regional Board within ninety (90) days after the average dry-weather waste flow for any month equals or exceeds 75 percent of the design capacity of the waste treatment and/or disposal facilities. The Discharger's senior administrative officer shall sign a letter which transmits that report and certifies that the policy making body is adequately informed about it. The report shall include:
 - a. Average daily flow for the month, the date on which the instantaneous peak flow occurred, the rate of that peak flow, and the total flow for the day.
 - b. The discharger's best estimate of when the average daily dry-weather flow rate will equal or exceed the design capacity of the treatment facilities.
 - c. The discharger's intended schedule for studies, design, and other steps needed to provide additional capacity for the waste treatment and/or disposal facilities before the waste flow rate equals the capacity of present units.

B. Self Monitoring Reports (SMRs)

1. At any time during the term of this permit, the State or Regional Water Board may notify the Discharger to electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). Until such notification is given, the Discharger shall submit hard copy SMRs. The CIWQS Web site will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under Sections III through IX. Additionally, the Discharger shall report in the SMR the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions – VI.C. of this Order. The Discharger shall submit monthly, quarterly, and annual SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table 11. Monitoring and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On	Monitoring Period	SMR Due Date
Continuous	The effective day of this Order	All	Submit with monthly SMR
Daily	"	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	Submit with monthly SMR
Weekly	"	Sunday through Saturday	Submit with monthly SMR
Monthly	"	1 st day of calendar month through last day of calendar month	first day of the second month following the reporting period, submit as monthly SMR
Quarterly	"	January 1 through March 31; samples shall be collected in January; April 1 through June 30; samples shall be collected in April; July 1 through September 30; samples shall be collected in July; October 1 through December 31; samples shall be collected in October;	first day of the second month following the reporting period, submit with monthly SMR
Semiannually	"	January 1 through June 30 July 1 through December 31	first day of the second month following the reporting period, submit with monthly SMR
Annually	"	January 1 through December 31	April 1 each year

4. Reporting Protocols. The Discharger shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR Part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The *estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc."). The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
 - d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from *extrapolation* beyond the lowest point of the calibration curve.
5. Multiple Sample Data. When determining compliance with an AMEL for priority pollutants and more than one sample result is available in a month, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
- a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
6. The Discharger shall submit hard copy SMRs (with an original signature) when required by subsection B.1 above in accordance with the following requirements:

- a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations.
 - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
 - c. SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the address listed below:

California Regional Water Quality Control Board
Santa Ana Region
3737 Main Street, Suite 500
Riverside, CA 92501-3348
7. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
8. By April 1 of each year, the discharger shall submit an annual report to the Regional Water Board. The annual report shall include the following:
- a. Tabular and graphical summaries of the monitoring data obtained during the previous year;
 - b. A discussion of the compliance record and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the waste discharge requirements;
 - c. A summary of the quality assurance (QA) activities for the previous year; and
 - d. For priority pollutant constituents that do not have effluent limitations but are required to be monitored, the Discharger shall evaluate the monitoring data obtained during the previous year and determine whether detected constituents are at levels that would warrant reopening the permit to include effluent limitations for such constituent(s). To conduct this evaluation, the concentration of detected constituents shall be compared to the most stringent applicable receiving water objectives (freshwater or human health (consumption of organisms only) as specified for that pollutant in 40 CFR 131.38²¹). The Discharger shall include a discussion of the corrective actions taken or planned to address values above receiving water objectives.

²¹ See *Federal Register*/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations

C. Discharge Monitoring Reports (DMRs)

1. As described in Section X.B.1 above, at any time during the term of this permit, the State or Regional Water Board may notify the Discharger to electronically submit SMRs that will satisfy federal requirements for submittal of Discharge Monitoring Reports (DMRs). Until such notification is given, the Discharger shall submit DMRs in accordance with the requirements described below.
2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharge shall submit the original DMR and one copy of the DMR to the address listed below:

Standard Mail	FedEx/UPS/ Other Private Carriers
State Water Resources Control Board Division of Water Quality c/o DMR Processing Center PO Box 100 Sacramento, CA 95812-1000	State Water Resources Control Board Division of Water Quality c/o DMR Processing Center 1001 I Street, 15 th Floor Sacramento, CA 95814

3. All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1). Forms that are self-generated or modified cannot be accepted.

Regional Administrator
U. S. Environmental Protection Agency
Region 9 – Attention WTR – 7
75 Hawthorne Street
San Francisco, CA 94105

D. Other Reports – Not Applicable

ATTACHMENT F – FACT SHEET

TABLE OF CONTENTS

Attachment F – Fact Sheet	F-3
I. Permit Information.....	F-3
II. Facility Description.....	F-4
A. Description of Wastewater and Biosolids Treatment or Controls	F-4
1. Facility Background	F-4
2. Design Characteristics.....	F-5
3. Biosolids/Sludge Handling Practices	F-5
B. Discharge Points and Receiving Waters.....	F-6
C. Summary of Previous Requirements and Self-Monitoring Report (SMR) Data.....	F-7
D. Compliance Summary.....	F-8
E. Planned Changes	F-10
III. Applicable Plans, Policies, and Regulations	F-10
A. Legal Authorities	F-10
B. California Environmental Quality Act (CEQA)	F-10
C. State and Federal Regulations, Policies, and Plans	F-10
D. Impaired Water Bodies on CWA 303(d) List-Not Applicable	F-13
E. Other Plans, Policies and Regulations-Not Applicable	F-13
IV. Rationale For Effluent Limitations and Discharge Specifications	F-13
A. Discharge Prohibitions	F-13
B. Technology-Based Effluent Limitations.....	F-13
1. Scope and Authority	F-13
2. Applicable Technology-Based Effluent Limitations	F-14
C. Water Quality-Based Effluent Limitations (WQBELs).....	F-14
1. Scope and Authority	F-14
2. Applicable Water Quality Criteria and Development of WQBELs	F-15
3. Determining the Need for WQBELs	F-18
4. WQBEL Calculations	F-19
5. Whole Effluent Toxicity (WET).....	F-21
D. Best Professional Judgment-Based Effluent Limitations.....	F-21
E. Summary of Final Effluent Limitations.....	F-21
1. Satisfaction of Anti-Backsliding Requirements.....	F-21
2. Satisfaction of Antidegradation Policy.....	F-21
3. Stringency of Requirements for Individual Pollutants.....	F-21
F. Interim Effluent Limitations – DP 001 and DP 002.....	F-22
G. Land Discharge Specifications – Not Applicable.....	F-23
H. Reclamation Specifications - DP 003.....	F-23
V. Rationale for Receiving Water Limitations	F-24
A. Surface Water	F-24
B. Groundwater	F-24
VI. Rationale for Monitoring and Reporting Requirements	F-24
A. Influent Monitoring	F-24
B. Effluent Monitoring.....	F-24

C.	Whole Effluent Toxicity Testing Requirements	F-25
D.	Receiving Water Monitoring	F-26
1.	Surface Water	F-26
2.	Groundwater – Not Applicable	F-26
E.	Other Monitoring Requirements	F-26
VII.	Rationale for Provisions	F-26
A.	Standard Provisions	F-26
B.	Special Provisions	F-27
VIII.	Public Participation	F-29
A.	Notification of Interested Parties	F-29
B.	Written Comments	F-29
C.	Public Hearing	F-29
D.	Waste Discharge Requirements Petitions	F-30
E.	Information and Copying	F-30
F.	Register of Interested Persons	F-30
G.	Additional Information	F-30

ATTACHMENT F – FACT SHEET

As described in Section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table 1. Facility Information

WDID	8 360112001
Discharger	City of Rialto
Name of Facility	City of Rialto Municipal Wastewater Treatment Plant
Address	501 E. Santa Ana Avenue
	Bloomington, CA 92316
	San Bernardino County
Facility Contact, Title and Phone	David Kachelski, Project Manager, Veolia Water North America West, LLC, City of Rialto, Phone: (909) 877-2752
Authorized Person to Sign and Submit Reports	David Kachelski, Project Manager, Veolia Water North America West, LLC, City of Rialto, Phone: (909) 877-2752, Derek Wieske, Deputy Director of Public Works 335 West Rialto Avenue, Rialto, CA 92376, (909) 820-2531
Mailing Address	501 E. Santa Ana Ave., Bloomington, CA 92316
Billing Address	335 West Rialto Ave., Rialto, CA 92376
Type of Facility	POTW
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	Y
Reclamation Requirements	Producer/User
Facility Permitted Flow	11.7 mgd
Facility Design Flow	11.7 mgd
Watershed	Santa Ana River Watershed
Receiving Water	Reach 4 of Santa Ana River; Groundwater Management Zones Riverside-A, Riverside-B and Chino 3¹
Receiving Water Type	Inland Surface Water and Groundwater

¹ The Chino 3 Groundwater Management Zone is part of the Chino North Groundwater Management Zone to which maximum benefit objectives for total dissolved solids and nitrate-nitrogen apply if the Chino Basin Watermaster (Watermaster) and the Inland Empire Utilities District (IEUA) fulfill a maximum benefit program specified in the Basin Plan. These maximum benefit objectives apply for the purpose of regulating the Discharger, in lieu of the Chino 3 “antidegradation” objectives, provided that the Watermaster/IEUA confirm that discharges by the Discharger are in conformance with their maximum benefit program.

- A.** The City of Rialto (hereinafter Discharger) is the owner of the City of Rialto Municipal Wastewater Treatment Plant (hereinafter Facility). The Discharger contracts with Veolia Water North America West, LLC to operate the Facility.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B.** The Facility discharges wastewater to Reach 4 of the Santa Ana River, a water of the United States. The discharge is currently regulated by Order No. 01-4, which was adopted on June 1, 2001 and expired on June 1, 2006. The terms and conditions of the current Order have been automatically continued and remain in effect until new Waste Discharge Requirements and NPDES permit are adopted pursuant to this Order.
- C.** The Discharger filed a report of waste discharge and submitted an application for renewal of its Waste Discharge Requirements (WDRs) and National Pollutant Discharge Elimination System (NPDES) permit on December 11, 2005. Supplemental Information was requested starting January 17, 2006. The latest supplemental information was received on December 20, 2006. A site visit was conducted on May 30, 2006, to observe operations and collect additional data to develop permit limitations and conditions. The application was deemed complete on December 20, 2006.

II. FACILITY DESCRIPTION

A. Description of Wastewater and Biosolids Treatment or Controls

1. Facility Background

This Facility is a domestic wastewater treatment plant with primary, secondary, and tertiary treatment. The Facility is located at 501 East Santa Ana Avenue in the Bloomington area of San Bernardino County.

The original wastewater treatment plant was constructed in 1956. Plant additions were constructed in 1965, 1981, 1990, 1994, and 1998. In 1998, construction of Phase I of the treatment plant expansion commenced. Phase I of the treatment plant expansion involved the construction of a new 4.7 million gallons per day (mgd) secondary treatment train (Plant No. 5) and construction of tertiary treatment facilities, including an 8 mgd UV disinfection system. Plant No. 5, including the new tertiary treatment facilities, started operation in the summer of 2000. This Facility provides wastewater treatment service within the City of Rialto, a portion of the City of Fontana and an area within the sphere of influence south of I-10 & east of Sierra Boulevard. The total population served is about 100,000.

2. Design Characteristics

The Facility includes five parallel treatment plants. Each of the five plants provides primary treatment and activated sludge treatment with facilities common to all five plants including preliminary treatment, tertiary treatment and solids handling. The treatment plant process is summarized as follows:

Table 2. City of Rialto MWTP

Individual Processes at Each Plant		Shared Processes		
Primary Treatment	Secondary Treatment	Preliminary Treatment	Tertiary Treatment	Solids Handling
Primary Clarifier	Activated Sludge Secondary Clarifiers	Bar Screens, Aerated Grit Chamber, Flow Equalization	Flow Equalization, Filtration, Chlorination, Dechlorination, UV Disinfection	Gravity Thickeners, Anaerobic Digestion Belt Press, Digester Gas Utilization Facility Sludge Drying Beds

The combined total treatment design capacity of Plants 1 through 5 is 11.7 mgd. Based on actual flow data from 2004, the maximum influent wet weather flow (recorded in October) was 13.2 mgd and average influent dry weather flow was 9 mgd.

3. Biosolids/Sludge Handling Practices

Solids wasted from each plant's secondary treatment processes are pumped to a gravity belt thickener to reduce the water content prior to being pumped to the anaerobic digester for stabilization. While there is no dedicated primary sludge thickener at the plant, solids concentration of primary sludge is maximized through careful monitoring and adjustment of primary sludge pumping controls.

Solids are stabilized by the anaerobic digestion process and then dewatered by a belt press. To reduce water content in the stabilized sludge, a polymer is injected into the sludge as it is being pumped to the belt press. The sludge/polymer mixture is subjected to 3 distinct de-watering zones of the belt press: the gravity zone, wedge zone, and the pressure zone. Typically, sludge pumped to the belt press is approximately 2 to 3 percent solids. Sludge leaving the belt press is approximately 14 to 16 percent solids. The filtrate removed from the de-watered sludge is pumped back to the liquid process flow streams for further treatment.

The plant utilizes a concrete-lined sludge drying bed where dewatered solids from the belt press may be spread by tractor for further solar drying.

Generally, during rainy seasons, de-watered solids from the belt press are placed directly into trucks and taken to a composting facility (One-Stop Composting) where the solids are blended with green waste and composted in accordance with biosolids

regulations. During the rest of the year, the drying bed is utilized prior to removal by the composting company.

Attachment B provides a map of the area around this Facility.
Attachment C provides a flow schematic for this Facility.

B. Discharge Points and Receiving Waters

1. Discharge Points for tertiary treated water

The Discharger discharges treated wastewater to surface waters at two points designated Discharge Point (DP) Serials No. 001 and 002, see Table 3. Discharges from DP 002 occur under emergency conditions only.

2. Discharge Point for recycling water reuse:

DP 003 is the discharge point for treated wastewater used for recycling purposes. Approximately 0.11 mgd (annual average flow) is used for Caltrans I-10 Freeway Irrigation from Pepper Avenue to Cherry Avenue.

3. Stormwater

Stormwater runoff from the facility site is channeled to a collection pit prior to pumping to the Rialto MWTP headworks for treatment. There are no discharges of stormwater runoff from the site into surface waters.

4. Receiving Waters

About 11.5 mgd of treated wastewater effluent flow from the plant is discharged via DPs 001 and/or 002 to Reach 4 of the Santa Ana River and about 0.11 mgd of effluent flow is diverted via DP 003 to Caltran's irrigation area. The Santa Ana River is a water of the U.S. The existing recycled water use area overlies the Riverside-B Groundwater Management Zone and Chino 3 "Antidegradation" Management Zone (See Table 3). (As noted in footnote 1 and IV.H. Reclamation Specifications – DP003, the Chino 3 Groundwater Management Zone is considered a part of the Chino North management zone when maximum benefit commitments by the Chino Basin Watermaster and the Inland Empire Utilities Agency are met.) The facility location overlies the Riverside A Groundwater Management Zone. It is possible that future recycled water use areas may be located in Riverside A, Rialto, and Colton Groundwater Management Zones.

Table 3. Summary of Discharge Points and receiving waters

Discharge Serial No.	Latitude	Longitude	Description and Receiving Waters	Flow & Frequency
001	34°03'04"N	117°21'11"W	Lined flood control channel tributary to Santa Ana River, Reach 4, which overlies the Riverside-A Groundwater Management Zone	11.5 mgd, continuous discharge
002	34°02'36"N	117°21'30"W	Emergency discharge point to the Santa Ana River (Reach 4), which overlies the Riverside-A Groundwater Management Zone	Emergency discharge

Table 3. Summary of Discharge Points and receiving waters

Discharge Serial No.	Latitude	Longitude	Description and Receiving Waters	Flow & Frequency
003	34°03'12"N	117°21'36"W	Recycled Water for Caltrans irrigation areas which overlie the Riverside-B and Chino 3 ² Groundwater Management Zones. Future recycled water use area- Riverside A	Currently up to 0.2 mgd Frequency varies

The Facility location map and treatment flow schematic are shown in Attachments B and C.

C. Summary of Previous Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations/Discharge Specifications contained in the previous Order No. 01-4 for discharges from Discharge Point 001 (Monitoring Location No. M-001) and representative monitoring data from the term of the previous Order are as follows:

Table 4. Historic Effluent Limitations and Monitoring Data

Parameter (units)	Effluent Limitation ¹			Monitoring Data (From 2003 – To 2006)			
	Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge	Highest 12-Month Average
pH Daily Average (SU)			6.5 – 8.5			7.8	
Chlorine Residual (mg/L)			0.1	<0.1		0.53	
BOD ₅ (mg/L)	20	30		6.1	23		
Suspended Solids (mg/L)	20	30		1.6	3.2		
Coliform Organisms (MPN/100 mL)	23 (1 per month)					30	
	2.2 (7-day median)					2.2	
Ammonia-Nitrogen (mg/L)	5.0			0.56			
TDS (mg/L)	400 (12-mo. avg)						487
Total Hardness (mg/L)	240 (12-mo. avg)						174
TIN (mg/L)	10 (12-mo. avg)						8.4
Selenium (µg/L)	4.1		8.2	3.2			
BHC gamma (µg/L)	0.063		0.127	< 0.021			
Nickel (µg/L)	60		121	8.7			
Chromium IV (µg/L)	8.1		16.3	5.0			
Copper, (µg/L)						39	
Cyanide (µg/L)						6	
Mercury (µg/L)						0.23	

² See footnote 1 and IV.H. Reclamation Specifications – DP003.

Table 4. Historic Effluent Limitations and Monitoring Data

Parameter (units)	Effluent Limitation ¹			Monitoring Data (From 2003 – To 2006)			
	Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge	Highest 12-Month Average
Bis(2-ethylhexyl) phthalate (µg/L)						15	
Chloride (mg/L)	35 (12-mo. avg)						97
Sodium (mg/L)	35 (12-mo. avg)						97
Sulfate (mg/L)	64 (12-mo. avg)						78

¹ Effluent limitations set forth by Order No. 01-4 (NPDES No. CA0105295), 2001.

D. Compliance Summary

Based on a review of effluent monitoring data submitted by the Discharger for the period from 2003 through 2006, the wastewater discharged from the wastewater treatment facility was in violation of the following effluent limitations:

Table 5. Compliance Summary

Date	Parameter	Value	Permit Limit	Reason for Violation	Corrective Measures
6/2/2003	Toxicity	>1.7 TUc	1.7 TUc	Samples failed 100% at all dilutions.	Accelerated testing performed. New samples passed test.
1/11/2004	Chlorine Residual	0.53 mg/L	0.1 mg/L	De-chlorination equipment (sulfonator) malfunction.	Staff placed sulfonator in manual mode and properly adjusted dosage.
2/24/2004	Turbidity	2.3 NTU	2 NTU (average - within any calendar day)	Solids escaping from Plant No. 3's secondary clarifier during peak hydraulic loading and a rain event	Mitigated the effects of the high turbidity event on downstream treatment processes.
4/3/2004	Coliform	2.2 MPN/ 100 mL	2.2 MPN/ 100mL (7-day median)	Interruption of bleach flow to the east side of the chlorine contact chamber due to instrumentation and human error.	Bleach flow to the east side of the chamber restored. Preventative maintenance activities carried out.
11/8/2004 11/10/2004 11/12/2004	Toxicity	> 1.7 TUc	1.7 TUc	Fontana Polishing and Plating in violation of pre-treatment permit.	Investigation of Fontana Polishing and Plating process. "Show Cause" hearing scheduled for February 2005.
1/10/2005	Turbidity	10.9 NTU	10 NTU (at any time)	Settled solids escaping from Plant 4's secondary clarifier caused by storm water related inflow/ infiltration.	Worked with City of Rialto and San Bernardino Flood Control.

Table 5. Compliance Summary

Date	Parameter	Value	Permit Limit	Reason for Violation	Corrective Measures
2/24/05	Coliform	< 2.2 MPN/100 mL	2.2 MPN/100mL (7-day median)	Filter cells No. 4 and 8 out of service due to mechanical failures.	Preventative maintenance activities carried out.
1/14/2006	CT	<450	Not <450 mg/min	Erratic operation of failing CL2 analyzer sample pump	Replaced failed CL2 analyzer pump on 2/1/2006
1/17/2006	CT	<450	Not <450 mg/min	Erratic operation of failing CL2 analyzer sample pump	Replaced failed CL2 analyzer pump on 2/1/2006
1/30/2006	CT	<450	Not <450 mg/min	Erratic operation of failing CL2 analyzer sample pump	Replaced failed CL2 analyzer pump on 2/1/2006
2/1/2006	Chlorine residual	2 mg/L 9 times in one hour period	>2 mg/L	Complete CL2 analyzer sample pump failure	Increased SBS in hand. Tested receiving waters. Replaced failed CL2 analyzer sample pump
3/8/2006	CT	<450	Not <450 mg/min	Mechanical failure. CL2 analyzer feed water pump broke (pump shaft sheared)	Installed temporary pump until replacement pump could be installed.
5/5/2006	CT	<450	Not <450 mg/min	Flow spike from gravity filter backwash	Noted and logged event.
6/4/2006	CT	<450	Not <450 mg/min	Unknown	Put bleach pumps in hand. Added HTH to contact chamber by hand. Increased CL2 dose.
7/12/2006	CT	<450	Not <450 mg/min	Possibly because of low DO in aeration caused by blower out of service	Increased bleach setpoint
7/25/2006	CT	<450	Not <450 mg/min	Low bleach setpoint	Increased bleach setpoint
8/7/2006	CT	<450	Not <450 mg/min	Caused by end of UV system test	Increased bleach setpoint and decreased filter influent pump speed
9/27/2006	CT	<450	Not <450 mg/min	Plant 4 aeration tank out of service for maintenance	Increased bleach setpoint and decreased filter influent pump speed
9/28/2006	Chlorine residual	>2 mg/L	No excursion >2 mg/L	Maintenance on a PLC caused SBS pumps to shut off	Immediately restarted pump through SCADA
12/26/2006	CT	<450	Not <450 mg/min	Primary return pumps left on manual for longer than anticipated time frame	Increased the bleach feed setpoint, reduced filter influent pump speed and directed greater flow to Plant 5
12/27/2006	CT	<450	Not <450 mg/min	Primary return pumps left on manual for longer than anticipated time frame	Increased the bleach feed setpoint, reduced filter influent pump speed and directed greater flow to Plant 5

E. Planned Changes

Within 12 months, the Discharger will be preparing a master plan that addresses both the aging equipment at the facility and the possible need to expand to accommodate planned land development within the City.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in the proposed Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (commencing with Section 13370). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, Chapter 4, Division 7 of the Water Code (commencing with Section 13260).

B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100 through 21177. This action also involves the re-issuance of waste discharge requirements for an existing facility that discharges treated wastewater to land and as such, is exempt from the provisions of California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301.

C. State and Federal Regulations, Policies, and Plans

1. **Water Quality Control Plans.** . The Regional Water Board adopted an updated Water Quality Control Plan for the Santa Ana Basin (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, State Water Resources Control Board (State Water Board) Resolution No. 88-63 (Sources of Drinking Water Policy) requires that, with certain exceptions, the Regional Water Board assign the municipal and domestic water supply use to water bodies. Based on the exception criteria specified in Resolution No. 88-63, the Regional Water Board excepted Reach 4 of the Santa Ana River and downstream reaches from the municipal and domestic supply beneficial use.

On January 22, 2004, the Regional Water Board adopted Resolution No. R8-2004-0001, amending the Basin Plan to incorporate revised boundaries for groundwater subbasins, now termed “management zones”, new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. The State Water Board and Office of Administrative Law (OAL) approved the N/TDS Amendment on September 30, 2004 and December 23, 2004, respectively. The surface water standards components of the N/TDS Amendment are awaiting EPA approval. Effluent limitations for TDS and TIN in this Order are based on applicable Total Inorganic Nitrogen (TIN) and TDS wasteload allocations, adopted and now in effect as part of the N/TDS Amendment. The TDS and TIN limits are at least as stringent as the limits in the prior Order.

As previously discussed, the Facility discharges wastewater from three discharge points. The designated beneficial uses of receiving waters affected by the discharge from the Facility are as follows:

Table 6. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 002	Reach 4 of Santa Ana River (and downstream Reaches)	<u>Present or Potential:</u> Groundwater recharge (GWR), water contact recreation (REC-1), non-contact water recreation (REC-2), warm freshwater habitat, and wildlife habitat (WILD) Excepted from Municipal and Domestic supply
001, 002	Riverside-A Groundwater management zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Riverside-B and Chino 3 “antidegradation” management zones ³	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
001, 002, 003	Downgradient groundwater management zones	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.

Requirements of this Order implement the Basin Plan.

3. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR, which incorporated the NTR criteria that were applicable in California. The CTR was amended on February 13, 2001. These rules contain water quality criteria for priority pollutants.
4. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters*,

³ See footnote 1 and IV. H. Reclamation Specifications – DP 003.

Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP).

The SIP became effective on April 28, 2000 with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000 with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005 that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.

5. **Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised State and Tribal water quality standards (WQS) become effective for CWA purposes (40 CFR 131.21, 65 FR 24641, April 27, 2000). Under the revised regulation (also known as the Alaska rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.
6. **Antidegradation Policy.** Section 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of section 131.12 and State Water Board Resolution No. 68-16.
7. **Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and federal regulations at title 40, Code of Federal Regulations⁴ section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.
8. **Monitoring and Reporting Requirements.** Section 122.48 of 40 CFR requires that all NPDES permits specify requirements for recording and reporting monitoring results. Sections 13267 and 13383 of the CWC authorize the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP) establishes monitoring and reporting requirements to implement federal and State requirements. This MRP is provided in Attachment E.

⁴ All further statutory references are to title 40 of the Code of Federal Regulations unless otherwise indicated.

D. Impaired Water Bodies on CWA 303(d) List-Not Applicable

E. Other Plans, Policies and Regulations-Not Applicable

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: section 122.44(a) requires that permits include applicable technology-based limitations and standards; and section 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

A. Discharge Prohibitions

The discharge prohibitions are based on the Federal Clean Water Act, Basin Plan, State Water Board's plans and policies, U.S. Environmental Protection Agency guidance and regulations, and previous permit Order No. 01-4 provisions and are consistent with the requirements set for other discharges regulated by NPDES permits adopted by the Regional Water Board.

B. Technology-Based Effluent Limitations

1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at section 122.44, title 40 of the Code of Federal Regulations, require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at Part 133 and/or Best Professional Judgment (BPJ) in accordance with Part 125, section 125.3.

Regulations promulgated in 40 CFR §125.3(a)(1) require technology-based effluent limitations for municipal Dischargers to be placed in waste discharge requirements based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTWs [defined in Section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the USEPA Administrator.

Based on this statutory requirement, USEPA developed secondary treatment regulations, which are specified in 40 CFR Part 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD₅), total suspended solids (TSS), and pH.

2. Applicable Technology-Based Effluent Limitations

This facility meets the technology-based regulations for the minimum level of effluent quality attainable by secondary treatment in terms of BOD₅, total suspended solids and removal rate as summarized in Table 6, below. These effluent limitations have been carried over from the previous Order for secondary treated wastewater discharge under conditions of 20:1 or more dilution provided by natural flow in the river.

Table 7. Summary of Technology-based Effluent Limitations under 20:1 or More Dilution

Constituent	Average Weekly (mg/L)	Average Monthly (mg/L)	Average Monthly Removal Rate %
Biochemical Oxygen Demand, 5-day 20°C	45	30	85
Total Suspended Solids	45	30	85

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

Section 301(b) of the CWA and section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

Section 122.44(d)(1)(i) mandates that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

2. Applicable Water Quality Criteria and Development of WQBELs

- a. The Basin Plan specifies narrative and numeric water quality objectives applicable to surface water as follows.

Table 8. Summary of Applicable Basin Plan Water Quality Objectives

Constituents	Basis for Limitations
Ammonia Nitrogen	Dissociates under certain conditions to the toxic un-ionized form. Thus nitrogen discharges to the Santa Ana River pose a threat to aquatic life and instream beneficial uses, as well as to the beneficial uses of affected groundwater. The Basin Plan specifies total ammonia and un-ionized ammonia objectives and an effluent limit of 4.5 mg/L for discharges to Santa Ana River - Reach 4.
Hydrogen Ion (pH)	Hydrogen Ion (pH) is a measure of Hydrogen Ion concentration in the water. A pH range of 6.5 to 8.5 for surface water discharges is specified.
Oil & Grease	Oil and related materials have a high surface tension and are not soluble in water, resulting in odors and visual impacts.
Total Chlorine Residual	Chlorine and its reaction product are toxic to aquatic life. To protect aquatic life, the Basin Plan specifies that for wastewater discharged into inland surface waters the chlorine residual should not exceed 0.1 mg/L
Total Dissolved Solids	High levels of TDS can adversely impact aquatic life. The TDS limit for surface water discharges is based on the amended Basin Plan objectives and wasteload allocation of 490 mg/L and surface water discharge flow at 12 MGD.
Total Inorganic Nitrogen	Nitrogen discharges to the Santa Ana River pose a threat to aquatic life and instream beneficial uses, as well as to the beneficial uses of affected groundwater. The TIN limit for surface water discharges is based on the amended Basin Plan wasteload allocation of 10 mg/L and surface water discharge flow at 12 MGD.

TDS and TIN: The TDS and TIN limitations are specified in the Order for both discharges to surface waters (DP 001 to Reach 4 of Santa Ana River) and for recycled water use. The proposed TDS/TIN limits for direct discharges into the River are based on the waste load allocations specified in Table 5-5 of the amended Basin Plan and shown in Table 9, below.

Table 9. Wasteload Allocations

POTW	TDS, mg/L	TIN, mg/L
Rialto	490	10

This Order also includes a TDS limit based on the quality of the water supplied to the service area plus a reasonable use increment for TDS of 250 mg/L. This reasonable use increment is discussed and authorized in the Basin Plan.

For surface water discharges, the more restrictive of the TDS wasteload allocation-based limit or the TDS limit based on water quality plus a reasonable use increment applies to discharges from the facility.

TDS and TIN limitations for recycled water use are described in H. Reclamation Specifications – DP003, below.

In accordance with 40 CFR Section 122.45(d), there may be instances in which the basis for a limit for a particular continuous discharge may be impracticable to be stated as a maximum daily, average weekly, or average monthly effluent limitation. The Regional Water Board has determined that it is not practicable to express TDS and TIN effluent limitations as average weekly and average monthly effluent limitations because the TDS and TIN objectives in the Basin Plan were established primarily to protect the underlying groundwater. Consequently, a 12-month average period is more appropriate.

- b. **CTR and SIP.** The National Toxics Rule, California Toxics Rule (CTR) and State Implementation Policy specify numeric objectives for toxic substances and the procedures whereby these objectives are to be implemented. The procedures include those used to conduct reasonable potential analysis to determine the need for effluent limitations for priority and non-priority pollutants.
- c. **Requirement to meet 2.2 total coliform bacteria limit in the effluent.** Article 3, Section 60305 of Title 22, Chapter 3, "Use of Recycled water for impoundments" of the California Code of Regulations specifies that recycled water used as a source of supply in a nonrestricted recreational impoundment shall be at all times an adequately disinfected, oxidized, coagulated, clarified, filtered wastewater (tertiary treated). The degree of treatment specified represents an approximately 5-log reduction in the virus content of the water. The California State Department of Health Services (CDHS) has determined that this degree of virus removal is necessary to protect the health of people using these impoundments for water contact recreation. The CDHS has developed wastewater disinfection guidelines ("Wastewater Disinfection for Health Protection", Department of Health Services, Sanitary Engineering Branch, February 1987) for discharges of wastewater to surface waters where water contact recreation (REC-1) is a beneficial use. The disinfection guidelines recommend the same treatment requirements for wastewater discharges to REC-1 waters as those stipulated in Title 22 for supply of recycled water to nonrestricted recreational impoundments, since the public health risks under both scenarios are analogous. The disinfection guidelines are based on sound science and are widely used as guidance to assure public health and beneficial use protection.

Santa Ana River, Reach 4 is not a “nonrestricted recreational impoundment,” nor is “recycled water⁵” being used as a supply source for the River pursuant to the definitions in Title 22. However, except during major storms, most of the flow in the River is composed of treated municipal wastewater discharges. The River is used for water contact recreation and, accordingly, is designated REC-1 (water contact beneficial use). People recreating in the River face an exposure similar to those coming in contact with recycled water in an impoundment. Therefore, to protect the water contact recreation beneficial use and to prevent nuisance and health risk, it is necessary and appropriate to require the same degree of treatment for wastewater discharges to the River as would be required for the use of recycled water in a nonrestricted recreational impoundment. Thus, this Order specifies requirements based on tertiary or equivalent treatment.

- d. Discharge of secondary treated and disinfected wastewater when the natural flows in the river provide at least 20:1 dilution:

The Department of Health Services has determined that public health and water contact recreation (REC-1) beneficial uses will be protected provided that at least 20:1 dilution of secondary treated and disinfected wastewater discharges by natural receiving waters is achieved (“Wastewater Disinfection Guidelines Feb. 1987; these guidelines are based on sound science and are widely used as guidance to assure public health and beneficial use protection). Based on best professional judgment of the effluent limitations necessary to prevent nuisance and to assure public health and REC-1 use protection, it is necessary and appropriate to implement these guidelines in this Order.

The Santa Ana River is not naturally perennial. In dry weather, flow in the Santa Ana River is comprised predominantly of effluent discharges from municipal wastewater treatment Facility (POTWs), and very little natural flow exists. Under storm conditions, 20:1 (natural receiving waters to effluent) or more dilution of the effluent by storm flows may be provided. These storm conditions may also threaten the operational safety of the wastewater treatment facility through influx of infiltrated storm flows into the sanitary sewer system. The discharge of secondary effluent when at least 20:1 dilution is provided by the receiving waters may be necessary to protect the integrity of the Facility.

Based on best professional judgment of the effluent limitations necessary to prevent nuisance and to assure public health and REC-1 use protection, and taking into consideration the need to protect the integrity of the treatment works, it is appropriate to implement these guidelines in this Order. Accordingly, this Order specifies requirements based on secondary treatment for surface water discharges under conditions of at least 20:1 dilution of the wastewater by natural receiving waters.

⁵

As defined in the Reclamation Criteria, recycled water means water which, as a result of treatment of domestic wastewater, is suitable for a direct beneficial use or a controlled use that would not otherwise occur.

3. Determining the Need for WQBELs

In accordance with Section 1.3 of the SIP, the Regional Water Board conducted a reasonable potential analysis (RPA) for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the Order. The Regional Water Board analyzed effluent data to determine if a pollutant in a discharge has the reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that have the reasonable potential to cause or contribute to an excursion above a water quality standard, numeric WQBELs are required. The RPA considers criteria from the CTR, and when applicable, water quality objectives specified in the Basin Plan.

Sufficient data are needed to conduct a complete RPA. If data are not sufficient, the Discharger will be required to gather the appropriate data for the Regional Water Board to conduct the RPA. Upon review of the data, and if the Regional Water Board determines that WQBELs are needed to protect the beneficial uses, the permit will be reopened for appropriate modification.

The RPA was performed for the priority pollutants for which effluent data were available. By review data provided by the Discharger, BHC Gamma (Lindane), Chromium VI, Nickel, and Selenium, which were regulated in the Order No. 01-4, were not determined to have reasonable potential to cause an excursion above any applicable priority pollutant criteria or objective. Therefore, this Order does not have effluent limitations for these pollutants. However, the RPA analysis shows that priority pollutants copper, cyanide, mercury, and Bis(2-ethylhexyl phthalate have reasonable potential to exceed water quality objectives. Consequently, effluent limitations for these constituents are included in this Order

Table 11. is a summary of the RPA evaluation for copper, cyanide, mercury, and Bis(2-ethylhexyl phthalate using monitoring data submitted by the Discharger.

Table 11. RPA Evaluation

Parameter	Unit	Effluent	CTR			Is Effluent Limit Required?		
		MEC	CMC	CCC	Human Health for consumption of organisms	CMC	CCC	Human Health
Copper	µg/L	39	52.5	33.7	--	No	Yes	--
Cyanide	µg/L	6	22	5	--	yes	yes	--
Mercury	µg/L	0.23			0.051			yes
Bis(2-ethylhexyl) phthalate	µg/L	15			5.9			yes

4. WQBEL Calculations

a. Total Chlorine Residual

For discharges under conditions of 20:1 or more dilution, the total chlorine residual effluent limitation in Section IV.A.4.a. is calculated using the equation specified in the State Implementation Policy Section 1.4 - Calculation of Effluent Limitation:

$ECA, \text{ mg/L} = C + D \times (C-B)$ where:

ECA= Effluent Concentration allowance/effluent limit

B= Ambient background concentration

C= Criterion/objective

D= Dilution Credit

$$ECA = 0.1 + 20 \times (0.1 - 0) = 2.1 \text{ mg/L}$$

- b. For priority pollutants, water quality based effluent limits are based on monitoring results and the calculation process outlined in Section 1.4 of the California Toxic Rule and the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays and Estuaries of California are summarized in the following Table 12.

Table 12 shows the permit limit calculations for total recoverable copper, cyanide, total recoverable mercury, Bis(2-ethylhexyl) phthalate.

Table 12. Criteria calculation based on California Toxics Rule for freshwater discharges

RIALTO MUNICIPAL WASTEWATER TREATMENT PLANT

PERMIT LIMIT CALCULATION AND DETERMINATION OF THE MOST APPROPRIATE ML VALUE CONSIDERING CV

unit in ug/l

Constituent	Caltoxics				CV = 0.6			LTA	Aquatic		Human		Permit Limit
	Freshwater		Human Health		Acute M	Chronic M	Objective/limits		Health Limits	Concentration Limit			
	CMC	CCC	H ₂ O+Org	Org only	Acute LTA	Chronic LTA							
total recoverable Copper	52.48	33.68			16.85	17.75	16.85	MDEL	AMEL	MDEL/AMEL	MDEL	AMEL	
Cyanide, free	22.00	5.20			7.06	2.74	2.74	8.52	4.25		8.5	4.2	
total recoverable Mercury				0.051						0.102	0.051	0.051	
Bis(2-ethylhexyl) phthalate				5.9						11.8	5.9	11.8	
												5.9	

1. Effluent hardness is based on median value of effluent flow to Santa Ana River as 154 mg/l.
2. For Order No. R8-2007-0006.

5. Whole Effluent Toxicity (WET)

This Order does not specify WET limits but requires chronic toxicity monitoring. The monitoring data indicated that during the past three years (2004-2006), the monthly trigger of 1 TUC has been exceeded three separate times. However, subsequent monthly chronic toxicity results immediately following the exceedance event did not exceed a TUC of 1.

D. Best Professional Judgment-Based Effluent Limitations

For tertiary treated wastewater, the BOD₅ and TSS concentration limits are based on Best Professional Judgment. The technology-based secondary treatment standards specify BOD₅ and TSS concentration limits that are less stringent.

Table 13. Tertiary Effluent BOD₅ and TSS Limits

Constituent	Average Weekly	Average Monthly
Biochemical Oxygen Demand	30 mg/L	20 mg/L
Suspended Solids	30 mg/L	20 mg/L

E. Summary of Final Effluent Limitations

1. Satisfaction of Anti-Backsliding Requirements

All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order.

2. Satisfaction of Antidegradation Policy

Discharges in conformance with the requirements of this Order will not result in a lowering of water quality and therefore conform to antidegradation requirements specified in Resolution No. 68-16, which incorporates the federal antidegradation policy at 40 CFR 131.12 where, as here, is it applicable.

3. Stringency of Requirements for Individual Pollutants

Water quality-based effluent limitations have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant water quality-based effluent limitations were derived from the CTR, the CTR is the applicable standard pursuant to section 131.38. The scientific procedures for calculating the individual water quality-based effluent limitations for priority pollutants are based on the CTR-SIP, which was approved by USEPA on May 18, 2000. Apart from certain surface water standards changes resulting from the N/TDS Basin Plan amendment that do not materially affect the quality requirements for the discharges regulated by this Order, all beneficial uses and water quality objectives contained in

the Basin Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

4. Summary of Final Effluent Limitations:

Table 14. Summary of Water Quality-based Final Effluent Limitations for Discharge Points 001 and 002

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Max Daily	Instantaneous Minimum	Instantaneous Maximum	
BOD ₅	mg/L	20	30	--	--	--	PO
Total Suspended Solids	mg/L	20	30	--	--	--	PO
Total Residual Chlorine	mg/L	--	--	--	--	0.1	BP
TDS	mg/L	490 (12-M avg)	--	--	--	--	PO, BP
Ammonia-Nitrogen	mg/L	4.5	--	--	--	--	PO, BP
Total Inorganic Nitrogen	mg/L	10 (12-M avg)	--	--	--	--	PO, BP
pH	Std. unit	--	--	--	6.5	8.5	PO, BP
Copper	µg/L	26.1		52.4			CTR, SIP
Cyanide	µg/L	4.2		8.5			CTR, SIP
Mercury	µg/L	0.051		0.102			CTR, SIP
Bis(2-ethylhexyl) phthalate	µg/L	5.9		11.8			CTR, SIP
Coliform	MPN	--	2.2 Median of last 7 days	--	--	--	PO, Title 22

Notes: PO = Previous Order, 01-4; BP= Basin Plan.

F. Interim Effluent Limitations – DP 001 and DP 002

The Discharger submitted a request to set up interim limitations for cyanide, total recoverable mercury, and Bis(2-ethylhexyl) phthalate. Based on SIP Section 2.1 - Compliance Schedules, the Discharger demonstrated that it is infeasible for the Discharger to achieve immediate compliance with cyanide, total recoverable mercury, and Bis(2-ethylhexyl) phthalate limits and requested a compliance schedule. This Order

includes interim effluent limits for these parameters, which shall be effective starting on March 2, 2007 and ending on March 1, 2008.

Table 15. Summary of Interim Effluent Limitations at Discharge Points 001 and 002

Parameter	Units	Effluent Limitations		Basis
		Average Monthly	Max Daily	
Cyanide	µg/L	24	24	SIP
Total Recoverable Mercury	µg/L	0.23	0.23	SIP
Bis(2-ethylhexyl) phthalate	µg/L	15	15	SIP

G. Land Discharge Specifications – Not Applicable

H. Reclamation Specifications - DP 003

1. Section 13523 of the California Water Code provides that a Regional Water Board, after consulting with and receiving the recommendations from the CDHS and any party who has requested in writing to be consulted, and after any necessary hearing, shall prescribe water reclamation requirements for water which is used or proposed to be used as recycled water, if, in the judgment of the Board, such requirements are necessary to protect the public health, safety, or welfare. Section 13523 further provides that such requirements shall include, or be in conformance with, the statewide uniform water recycling criteria established by the CDHS pursuant to California Water Code Section 13521.
2. Reclamation specifications in the proposed Order are upon the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations, and the California Water Code Section 13521.
3. TDS/TIN limitations for recycled water reuse are based on the TDS/nitrate-nitrogen water quality objectives in Table 4-1 of the Basin Plan. A nitrogen loss coefficient of 25% is applied to the objectives in calculating the TIN limits, per the Basin Plan.

Table 16. TDS/TIN Limitations for Recycled Water Use

Groundwater Management Zone	Groundwater TDS, (mg/L)	TIN (mg/L)
Riverside-A	560	8.3
Riverside-B	290	10.1
Chino 3—Anti-degradation	260	4.7
Chino North-Max benefit	420	6.7

As noted in footnote 1, the Chino 3 Groundwater Management Zone is a part of the Chino North Management Zone to which maximum benefit TDS and nitrate-nitrogen objectives apply provided that the Chino Basin Watermaster (CBW) and the Inland

Empire Utilities Agency (IEUA) meet maximum benefit commitments that are specified in the Basin Plan. These maximum benefit objectives apply for the purposes of regulating recycled water use by the Discharger in areas overlying the Chino 3/Chino North Management Zone, in lieu of the Chino 3 antidegradation TDS and nitrate-nitrogen objectives, provided that the Discharger obtains confirmation from CBW/IEUA that the discharges are consistent with the CBW/IEUA maximum benefit program.

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water

1. The surface water receiving water limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plan.

B. Groundwater

1. The receiving groundwater limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plan.

VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

Section 122.48 of 40 CFR requires all NPDES permits to specify recording and reporting of monitoring results. Sections 13267 and 13383 of the CWC authorize the Water Boards to require technical and monitoring reports. The MRP, Attachment E of this Order, establishes monitoring and reporting requirements to implement federal and State requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

A. Influent Monitoring

This Order carries forward the treatment plant influent monitoring requirements without change. Influent monitoring is required to determine the effectiveness of the treatment program and assess treatment plant performance.

B. Effluent Monitoring

The Discharger is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the proposed monitoring and reporting program (Attachment E). This provision requires compliance with the monitoring and reporting program, and is based on 40 CFR 122.44(i), 122.62, 122.63 and 124.5. The SMP is a standard requirement in almost all NPDES permits (including the proposed Order) issued by the Regional Water Board. In addition to containing definitions of terms, it specifies general sampling/analytical protocols and the requirements of reporting of spills, violations, and routine monitoring data in accordance with NPDES regulations, the California Water Code, and Regional

Water Board's policies. The monitoring and reporting program also contains sampling program specific for the Discharger's wastewater treatment plant. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with Section 1.3 of the SIP, periodic monitoring is required for all priority pollutants defined by the CTR, for which criteria apply and for which no effluent limitations have been established, to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

This Order modifies the monitoring requirements specified in the Order No. 01-4 and adds monitoring requirements for EPA priority pollutants. This Order also requires the Discharger to conduct accelerated monitoring for those constituents that are detected in the annual priority pollutant scan.

C. Whole Effluent Toxicity Testing Requirements

Whole effluent toxicity (WET) protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. WET tests measure the degree of response of exposed aquatic test organisms to an effluent. The WET approach implements the narrative "no toxics in toxic amounts" criterion. There are two types of WET tests: acute and chronic. An acute toxicity test is conducted over a shorter time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth.

The Basin Plan specifies a narrative objective for toxicity, requiring that all waters be maintained free of toxic substances in concentrations that are lethal to or produce other detrimental response on aquatic organisms. Detrimental response includes but is not limited to decreased growth rate, decreased reproductive success of resident or indicator species, and/or significant alterations in population, community ecology, or receiving water biota.

In addition to the Basin Plan requirements, Section 4 of the SIP states that a chronic toxicity effluent limitation is required in permits for all discharges that will cause, have the reasonable potential to cause, or contribute to chronic toxicity in receiving waters. Therefore, in accordance with the SIP, this Order requires the Discharger to conduct chronic toxicity testing. In addition, the Order establishes thresholds that when exceeded requires the Discharger to conduct accelerated toxicity testing and/or conduct toxicity identification evaluation (TIE) studies.

This Order requires the Discharger to conduct chronic toxicity testing of the effluent on a monthly basis. The Order also requires the Discharger to conduct an Initial Investigation Toxicity Reduction Evaluation (IITRE) program when either the two-month median of toxicity test results exceeds 1 TUC or any single test exceeds 1.7 TUC for survival endpoint. Based on the results of this investigation program and at the discretion of the Executive Officer, a more rigorous Toxicity Reduction Evaluation/Toxicity Identification Evaluation (TRE/TIE) may be required. A re-opener

provision is included in the Order to incorporate a chronic toxicity effluent limitation if warranted by the toxicity test results.

D. Receiving Water Monitoring

1. Surface Water

- a. For discharges of treated and disinfected effluent when 20:1 or more dilution is provided by the creek at the point of discharge, the Order requires the Discharger to establish a sampling station(s) at a suitable location(s) where the flow⁶ in the Creek at the point of discharge can be determined. The Order also requires that flow measurements in the Creek are made prior to any direct discharge to the Creek and shall continue on a daily basis until the discharge is terminated.
- b. Receiving water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water. Requirements are based on the Basin Plan.

2. Groundwater – Not Applicable

E. Other Monitoring Requirements

1. **Water Supply Monitoring** - The Discharger is required to collect a sample of each source of water supplied and analyze for total dissolved solids. The result of this monitoring will enable the Discharger to show compliance with TDS limitations in the Order.
2. **Biosolids Monitoring** - This Order continues the monitoring requirements specified in Order No. 01-4, with minor modification. The Discharger is now required to submit monitoring data annually instead of quarterly.
3. **Pretreatment Monitoring** - These monitoring and reporting requirements are established pursuant EPA 40 CFR 403 regulations.

VII. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which in accordance with 40 CFR §§122.41 and 122.42, apply to all NPDES discharges and must be included in every NPDES permit, are provided in Attachment D to the Order.

Title 40 CFR Section 122.41(a)(1) and (b) through (n) establish conditions that apply to all State-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation

⁶ *Exclusive of discharges to surface waters from upstream publicly owned treatment works.*

to the regulations must be included in the Order. 40 CFR Section 123.25(a)(12) allows the State to omit or modify conditions to impose more stringent requirements. In accordance with Section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR Sections 122.41(j)(5) and (k)(2) because the enforcement authority under the CWC is more stringent. In lieu of these conditions, this Order incorporates by reference CWC Section 13387(e).

B. Special Provisions

1. Reopener Provisions

The provisions are based on 40 CFR Parts 122.44(c) and 123. The Regional Water Board may reopen the permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board or Regional Water Board, including revisions to the Basin Plan.

2. Special Studies and Additional Monitoring Requirements

- a. **Toxicity Identification Evaluations or Toxicity Reduction Evaluations.** This provision is based on the SIP, Section 4, Toxicity Control Provisions.
- b. **Antidegradation Analysis.** This provision is based on State Water Board Resolution No. 68-16, which requires the Board in regulating the discharge of waste to maintain high quality waters of the state (the Discharger must demonstrate that it has implemented adequate controls (e.g., adequate treatment capacity) to ensure that high quality waters will be maintained).
- c. To assure that discharges occur only when the receiving waters can provide 20:1 dilution, the Order requires the Discharger to make provisions for the measurement of the receiving water flow at a suitable location and determine whether 20:1 dilution exists at the point of discharge before discharging secondary treated effluent.

3. Best Management Practices and Pollution Prevention - The requirements are based on the SIP Section 2.4.5.1

4. Construction, Operation, and Maintenance Specifications - The requirements are based on requirements that were specified in the prior Order.

5. Special Provisions for Municipal Facility - POTWs Only

- a. **Biosolids:** On February 19, 1993, the USEPA issued a final rule for the use and disposal of sewage sludge, 40 CFR, Part 503. This rule requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. The State of California has not been delegated the authority to implement this program, therefore, the U.S. Environmental Protection Agency is the implementing agency.

- b. Oxidized, filtered, and disinfected by UV and/or chlorine Wastewater Requirements: These requirements are based on Title 22 requirements for the use of recycled water.
- c. Pretreatment: The treatment plant capacity is 11.7 mgd and there are significant industrial users within the service areas. Consequently, this Order contains requirements for the implementation of an effective pretreatment program pursuant to Section 307 of the Federal Clean Water Act; Parts 35 and 403 of Title 40, Code of Federal Regulations (40 CFR 35 and 40 CFR 403); and/or Section 2233, Title 23, California Code of Regulations.

6. Other Special Provisions – Not Applicable

7. Compliance Schedules

This Order establishes final effluent limitations for cyanide, Bis(2-ethylhexyl) phthalate, and mercury that are new limits for the discharges. This Order also contains a compliance schedule that provides the Discharger time to bring their discharges into compliance with the newly established final limits. In accordance with Section 2.1 of the SIP, compliance schedules can only be provided by the Board after the Discharger has submitted a report that demonstrates that it is infeasible for the Discharger to achieve immediate compliance with newly established final effluent limitations.

On December 20, 2006, the Discharger requested that a compliance schedule for cyanide, Bis(2-ethylhexyl) phthalate, and mercury limitations be included in this Order. The Discharger demonstrated that immediate compliance with the proposed effluent limitations for cyanide, Bis(2-ethylhexyl) phthalate, and mercury is not feasible.

The compliance schedule included in this Order is based on the shortest practicable time required to achieve compliance, but it exceeds one year. Consequently, this Order includes interim as well as final limits and a schedule for compliance with the final cyanide, Bis(2-ethylhexyl) phthalate, and mercury limitations. The interim average monthly effluent limits are based on the maximum detected effluent concentration for cyanide, Bis(2-ethylhexyl) phthalate, and mercury at this Facility. This is in accordance with SIP Section 2.2.1, which stipulates that "Numeric interim limitations for the pollutant must be based on current treatment facility performance or on previous permit limitations, whichever is more stringent." Since there is no limit on cyanide, Bis(2-ethylhexyl) phthalate, and mercury in the prior Order No. 01-4, interim limits must be based on the current treatment plant performance.

The proposed permit allows the Discharger up to March 1, 2008 to achieve compliance with the final cyanide, Bis(2-ethylhexyl) phthalate, and mercury limitations. Quarterly status reporting is required to inform the Regional Water Board about the progress made by the Discharger to achieve compliance with the final limitations within the specified time.

VIII. PUBLIC PARTICIPATION

The Regional Water Board is considering the issuance of waste discharge requirements (WDRs) that will serve as a National Pollutant Discharge Elimination System (NPDES) permit for the Municipal Wastewater Treatment Plant. As a step in the WDR adoption process, the Regional Water Board staff has developed tentative WDRs. The Regional Water Board encourages public participation in the WDR adoption process.

A. Notification of Interested Parties

The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through the posting of Notice of Public Hearing at the Rialto City Hall and at the local newspaper; and at the Regional Water Board website: <http://www.waterboards.ca.gov/santaana> on February 2, 2007.

B. Written Comments

The staff determinations are tentative. Interested persons are invited to submit written comments concerning these tentative WDRs. Comments must be submitted either in person or by mail to the Executive Office at the Regional Water Board at the address above on the cover page of this Order.

To be fully responded to by staff and considered by the Regional Water Board, written comments should be received at the Regional Water Board offices by 5:00 p.m. on February 12, 2007.

Jane Qiu
California Regional Water Quality Control Board
Santa Ana Region
3737 Main Street, Suite 500
Riverside, CA 92501-3348

C. Public Hearing

The Regional Water Board will hold a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: March 2, 2007
Time: 9:00 A.M.
Location: City of Council Chambers of Loma Linda
25541 Barton Road
City of Loma Linda, CA

Interested persons are invited to attend. At the public hearing, the Regional Water Board will hear testimony, if any, pertinent to the discharge, WDRs, and permit. Oral testimony will be heard; however, for accuracy of the record, important testimony should be in writing.

Please be aware that dates and venues may change. Our web address <http://www.waterboards.ca.gov/santaana> where you can access the current agenda for changes in dates and locations.

D. Waste Discharge Requirements Petitions

Any aggrieved person may petition the State Water Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be submitted within 30 days of the Regional Water Board's action to the following address:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

E. Information and Copying

The Report of Waste Discharge (RWD), related documents, tentative effluent limitations and special provisions, comments received, and other information are on file and may be inspected at the address above at any time between 9:00 a.m. and 3:00 p.m. Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (951) 320-2008.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this Order should be directed to Jane Qiu at (951) 320-2008.

ATTACHMENT G - EPA PRIORITY POLLUTANT LIST

EPA PRIORITY POLLUTANT LIST		
Metals	Acid Extractibles	Base/Neutral Extractibles (continuation)
1. Antimony	45. 2-Chlorophenol	91. Hexachloroethane
2. Arsenic	46. 2,4-Dichlorophenol	92. Indeno (1,2,3-cd) Pyrene
3. Beryllium	47. 2,4-Dimethylphenol	93. Isophorone
4. Cadmium	48. 2-Methyl-4,6-Dinitrophenol	94. Naphthalene
5a. Chromium (III)	49. 2,4-Dinitrophenol	95. Nitrobenzene
5b. Chromium (VI)	50. 2-Nitrophenol	96. N-Nitrosodimethylamine
6. Copper	51. 4-Nitrophenol	97. N-Nitrosodi-N-Propylamine
7. Lead	52. 3-Methyl-4-Chlorophenol	98. N-Nitrosodiphenylamine
8. Mercury	53. Pentachlorophenol	99. Phenanthrene
9. Nickel	54. Phenol	100. Pyrene
10. Selenium	55. 2, 4, 6 – Trichlorophenol	101. 1,2,4-Trichlorobenzene
11. Silver	Base/Neutral Extractibles	Pesticides
12. Thallium	56. Acenaphthene	102. Aldrin
13. Zinc	57. Acenaphthylene	103. Alpha BHC
Miscellaneous	58. Anthracene	104. Beta BHC
14. Cyanide	59. Benzidine	105. Delta BHC
15. Asbestos (not required unless requested)	60. Benzo (a) Anthracene	106. Gamma BHC
16. 2,3,7,8-Tetrachlorodibenzo-P-Dioxin (TCDD)	61. Benzo (a) Pyrene	107. Chlordane
Volatile Organics	62. Benzo (b) Fluoranthene	108. 4, 4' - DDT
17. Acrolein	63. Benzo (g,h,i) Perylene	109. 4, 4' - DDE
18. Acrylonitrile	64. Benzo (k) Fluoranthene	110. 4, 4' - DDD
19. Benzene	65. Bis (2-Chloroethoxy) Methane	111. Dieldrin
20. Bromoform	66. Bis (2-Chloroethyl) Ether	112. Alpha Endosulfan
21. Carbon Tetrachloride	67. Bis (2-Chloroisopropyl) Ether	113. Beta Endosulfan
22. Chlorobenzene	68. Bis (2-Ethylhexyl) Phthalate	114. Endosulfan Sulfate
23. Chlorodibromomethane	69. 4-Bromophenyl Phenyl Ether	115. Endrin
24. Chloroethane	70. Butylbenzyl Phthalate	116. Endrin Aldehyde
25. 2-Chloroethyl Vinyl Ether	71. 2-Chloronaphthalene	117. Heptachlor
26. Chloroform	72. 4-Chlorophenyl Phenyl Ether	118. Heptachlor Epoxide
27. Dichlorobromomethane	73. Chrysene	119. PCB 1016
28. 1,1-Dichloroethane	74. Dibenzo (a,h) Anthracene	120. PCB 1221
29. 1,2-Dichloroethane	75. 1,2-Dichlorobenzene	121. PCB 1232
30. 1,1-Dichloroethylene	76. 1,3-Dichlorobenzene	122. PCB 1242
31. 1,2-Dichloropropane	77. 1,4-Dichlorobenzene	123. PCB 1248
32. 1,3-Dichloropropylene	78. 3,3'-Dichlorobenzidine	124. PCB 1254
33. Ethylbenzene	79. Diethyl Phthalate	125. PCB 1260
34. Methyl Bromide	80. Dimethyl Phthalate	126. Toxaphene
35. Methyl Chloride	81. Di-n-Butyl Phthalate	
36. Methylene Chloride	82. 2,4-Dinitrotoluene	
37. 1,1,2,2-Tetrachloroethane	83. 2,6-Dinitrotoluene	
38. Tetrachloroethylene	84. Di-n-Octyl Phthalate	
39. Toluene	85. 1,2-Diphenylhydrazine	
40. 1,2-Trans-Dichloroethylene	86. Fluoranthene	
41. 1,1,1-Trichloroethane	87. Fluorene	
42. 1,1,2-Trichloroethane	88. Hexachlorobenzene	
43. Trichloroethylene	89. Hexachlorobutadiene	
44. Vinyl Chloride	90. Hexachlorocyclopentadiene	

ATTACHMENT H – MINIMUM LEVELS

MINIMUM LEVELS IN PPB (µg/l)

Table 1- VOLATILE SUBSTANCES ¹	GC	GCMS
Acrolein	2.0	5
Acrylonitrile	2.0	2
Benzene	0.5	2
Bromoform	0.5	2
Carbon Tetrachloride	0.5	2
Chlorobenzene	0.5	2
Chlorodibromomethane	0.5	2
Chloroethane	0.5	2
Chloroform	0.5	2
Dichlorobromomethane	0.5	2
1,1 Dichloroethane	0.5	1
1,2 Dichloroethane	0.5	2
1,1 Dichloroethylene	0.5	2
1,2 Dichloropropane	0.5	1
1,3 Dichloropropylene (volatile)	0.5	2
Ethylbenzene	0.5	2
Methyl Bromide (<i>Bromomethane</i>)	1.0	2
Methyl Chloride (<i>Chloromethane</i>)	0.5	2
Methylene Chloride (<i>Dichloromethane</i>)	0.5	2
1,1,2,2 Tetrachloroethane	0.5	1
Tetrachloroethylene	0.5	2
Toluene	0.5	2
trans-1,2 Dichloroethylene	0.5	1
1,1,1 Trichloroethane	0.5	2
1,1,2 Trichloroethane	0.5	2
Trichloroethylene	0.5	2
Vinyl Chloride	0.5	2
1,2 Dichlorobenzene (volatile)	0.5	2
1,3 Dichlorobenzene (volatile)	0.5	2
1,4 Dichlorobenzene (volatile)	0.5	2

Selection and Use of Appropriate ML Value:

ML Selection: When there is more than one ML value for a given substance, the discharger may select any one of those ML values, and their associated analytical methods, listed in this Attachment that are below the calculated effluent limitation for compliance determination. If no ML value is below the effluent limitation, then the discharger shall select the lowest ML value, and its associated analytical method, listed in the PQL Table.

ML Usage: The ML value in this Attachment represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interferences. Assuming that all method-specific analytical steps are followed, the ML value will also represent, after the appropriate application of method-specific factors, the lowest standard in the calibration curve for that specific analytical technique. Common analytical practices sometimes require different treatment of the sample relative to calibration standards.

Note: chemical names in parenthesis and italicized is another name for the constituent.

¹ The normal method-specific factor for these substances is 1, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance.

MINIMUM LEVELS IN PPB (µg/l)

Table 2 – Semi-Volatile Substances ²	GC	GCMS	LC
2-Chloroethyl vinyl ether	1	1	
2 Chlorophenol	2	5	
2,4 Dichlorophenol	1	5	
2,4 Dimethylphenol	1	2	
4,6 Dinitro-2-methylphenol	10	5	
2,4 Dinitrophenol	5	5	
2- Nitrophenol		10	
4- Nitrophenol	5	10	
4 Chloro-3-methylphenol	5	1	
2,4,6 Trichlorophenol	10	10	
Acenaphthene	1	1	0.5
Acenaphthylene		10	0.2
Anthracene		10	2
Benzidine		5	
Benzo (a) Anthracene (1,2 Benzanthracene)	10	5	
Benzo(a) pyrene (3,4 Benzopyrene)		10	2
Benzo (b) Flouranthene (3,4 Benzofluoranthene)		10	10
Benzo(g,h,i)perylene		5	0.1
Benzo(k)fluoranthene		10	2
bis 2-(1-Chloroethoxyl) methane		5	
bis(2-chloroethyl) ether	10	1	
bis(2-Chloroisopropyl) ether	10	2	
bis(2-Ethylhexyl) phthalate	10	5	
4-Bromophenyl phenyl ether	10	5	
Butyl benzyl phthalate	10	10	
2-Chloronaphthalene		10	
4-Chlorophenyl phenyl ether		5	
Chrysene		10	5
Dibenzo(a,h)-anthracene		10	0.1
1,2 Dichlorobenzene (semivolatile)	2	2	
1,3 Dichlorobenzene (semivolatile)	2	1	
1,4 Dichlorobenzene (semivolatile)	2	1	
3,3' Dichlorobenzidine		5	
Diethyl phthalate	10	2	
Dimethyl phthalate	10	2	
di-n-Butyl phthalate		10	
2,4 Dinitrotoluene	10	5	
2,6 Dinitrotoluene		5	
di-n-Octyl phthalate		10	
1,2 Diphenylhydrazine		1	
Fluoranthene	10	1	0.05
Fluorene		10	0.1
Hexachloro-cyclopentadiene	5	5	
1,2,4 Trichlorobenzene	1	5	

MINIMUM LEVELS IN PPB (µg/l)

Table 2 - SEMI-VOLATILE SUBSTANCES ²	GC	GCMS	LC	COLOR
Pentachlorophenol	1	5		
Phenol ³	1	1		50
Hexachlorobenzene	5	1		
Hexachlorobutadiene	5	1		
Hexachloroethane	5	1		
Indeno(1,2,3,cd)-pyrene		10	0.05	
Isophorone	10	1		
Naphthalene	10	1	0.2	
Nitrobenzene	10	1		
N-Nitroso-dimethyl amine	10	5		
N-Nitroso -di n-propyl amine	10	5		
N-Nitroso diphenyl amine	10	1		
Phenanthrene		5	0.05	
Pyrene		10	0.05	

Table 3- INORGANICS ⁴	FAA	GFAA	ICP	ICPMS	SPGFAA	HYDRIDE	CVAA	COLOR	DCP
Antimony	10	5	50	0.5	5	0.5			1000
Arsenic		2	10	2	2	1		20	1000
Beryllium	20	0.5	2	0.5	1				1000
Cadmium	10	0.5	10	0.25	0.5				1000
Chromium (total)	50	2	10	0.5	1				1000
Chromium VI	5							10	
Copper	25	5	10	0.5	2				1000
Lead	20	5	5	0.5	2				10000
Mercury				0.5			0.2		
Nickel	50	5	20	1	5				1000
Selenium		5	10	2	5	1			1000
Silver	10	1	10	0.25	2				1000
Thallium	10	2	10	1	5				1000
Zinc	20		20	1	10				1000
Cyanide								5	

² With the exception of phenol by colorimetric technique, the normal method-specific factor for these substances is 1000, therefore, the lowest standards concentration in the calibration curve is equal to the above ML value for each substance multiplied by 1000.

³ Phenol by colorimetric technique has a factor of 1.

⁴ The normal method-specific factor for these substances is 1, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance.

MINIMUM LEVELS IN PPB (µg/l)

Table 4- PESTICIDES – PCBs ⁵	GC
Aldrin	0.005
alpha-BHC (<i>a</i> -Hexachloro-cyclohexane)	0.01
beta-BHC (<i>b</i> -Hexachloro-cyclohexane)	0.005
Gamma-BHC (<i>g</i> -Hexachloro-cyclohexane) (Lindane; <i>g</i> -Hexachloro-cyclohexane)	0.02
Delta-BHC (<i>d</i> -Hexachloro-cyclohexane)	0.005
Chlordane	0.1
4,4'-DDT	0.01
4,4'-DDE	0.05
4,4'-DDD	0.05
Dieldrin	0.01
Alpha-Endosulfan	0.02
Beta-Endosulfan	0.01
Endosulfan Sulfate	0.05
Endrin	0.01
Endrin Aldehyde	0.01
Heptachlor	0.01
Heptachlor Epoxide	0.01
PCB 1016	0.5
PCB 1221	0.5
PCB 1232	0.5
PCB 1242	0.5
PCB 1248	0.5
PCB 1254	0.5
PCB 1260	0.5
Toxaphene	0.5

Techniques:

GC - Gas Chromatography

GCMS - Gas Chromatography/Mass Spectrometry

HRGCMS - High Resolution Gas Chromatography/Mass Spectrometry (i.e., EPA 1613, 1624, or 1625)

LC - High Pressure Liquid Chromatography

FAA - Flame Atomic Absorption

GFAA - Graphite Furnace Atomic Absorption

HYDRIDE - Gaseous Hydride Atomic Absorption

CVAA - Cold Vapor Atomic Absorption

ICP - Inductively Coupled Plasma

ICPMS - Inductively Coupled Plasma/Mass Spectrometry

SPGFAA - Stabilized Platform Graphite Furnace Atomic Absorption (i.e., EPA 200.9)

DCP - Direct Current Plasma

COLOR - Colorimetric

⁵ The normal method-specific factor for these substances is 100, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance multiplied by 100.

ATTACHMENT I – TRIGGERS FOR MONITORING PRIORITY POLLUTANTS

	CONSTITUENT	µg/L
1	Antimony	7
2	Arsenic	75
3	Beryllium	--
4	Cadmium	4.2
5a	Chromium III	147
5b	Chromium VI	5.7
6	Copper	16.8
7	Lead	12
8	Mercury	0.026
9	Nickel	37.5
10	Selenium	2.5
11	Silver	4.3
12	Thallium	3.2
13	Zinc	86
14	Cyanide	2.6
15	Asbestos	--
16	2,3,7,8-TCDD (Dioxin)	0.000000007
17	Acrolein	160
18	Acrylonitrile	0.03
19	Benzene	0.6
20	Bromoform	2.2
21	Carbon Tetrachloride	0.13
22	Chlorobenzene	340
23	Chlorodibromomethane	0.22
24	Chloroethane	--
25	2-Chloroethyl vinyl ether	--
26	Chloroform	--
27	Dichlorobromomethane	0.28
28	1,1-Dichloroethane	5
29	1,2-Dichloroethane	0.19
30	1,1-Dichloroethylene	0.029
31	1,2-Dichloropropane	0.26
32	1,3-Dichloropropylene	5
33	Ethylbenzene	0.3
34	Methyl Bromide	24
35	Methyl Chloride	--
36	Methylene Chloride	2.4
37	1,1,2,2-Tetrachloroethane	0.085

	CONSTITUENT	µg/L
38	Tetrachloroethylene	0.4
39	Toluene	0.15
40	1,2,-Trans-dichloroethylene	10
41	1,1,1-Trichloroethane	200
42	1,1,2-Trichloroethane	0.3
43	Trichloroethylene	1.35
44	Vinyl Chloride	0.5
45	2-Chlorophenol	60
46	2,4-Dichlorophenol	46.5
47	2,4-Dimethylphenol	270
48	2-Methy-4,6-Dinitrophenol	6.7
49	2,4-Dinitrophenol	35
50	2-Nitrophenol	--
51	4-Nitrophenol	--
52	3-Methyl-4-Chlorophenol	--
53	Pentachlorophenol	0.14
54	Phenol	10500
55	2,4,6-Trichlorophenol	1.05
56	Acenaphthene	600
57	Acenaphthylene	--
58	Anthracene	4800
59	Benzidine	0.00006
60	Benzo (a) anthracene	0.0022
61	Benzo (a) pyrene	0.0022
62	Benzo (b) fluoranthene	0.0022
63	Benzo (g,h,i) pyrene	--
64	Benzo (k) fluoranthene	0.0022
65	Bis (2-Chloroethoxy) methane	--
66	Bis (2-Chloroethyl) ether	0.016
67	Bis (2-Chloroisopropyl) ether	700
68	Bis (2-ethyhexyl) phthalate	0.9
69	4-Bromophenyl phenyl ether	--
70	Butyl benzyl phthalate	1500
71	2- Chloronaphthalene	850
72	4-Chlorophenyl phenyl ether	--
73	Chrysene	0.0022
74	Dibenzo (a,h) anthracene	0.0022
75	1,2-Dichlorobenzene	0.6

See notes below for italicized constituents.

ATTACHMENT I. -Continued

	CONSTITUENT	µg/L
76	1,3-Dichlorobenzene	200
77	<i>1,4-Dichlorobenzene</i>	<i>5</i>
78	3,3-Dichlorobenzidine	0.02
79	Diethyl phthalate	11,500
80	Dimethyl phthalate	156,500
81	Di-N-butyl phthalate	1,350
82	2,4-Dinitrotoluene	0.055
83	2,6-Dinitrotoluene	--
84	Di-N-octyl phthalate	--
85	1,2-Diphenylhydrazine	0.02
86	Fluoranthene	150
87	Fluorene	650
88	Hexachlorobenzene	0.00038
89	Hexachlorobutadiene	0.22
90	<i>Hexachlorocyclopentadiene</i>	<i>50</i>
91	Hexachloroethane	0.95
92	Indeno (1,2,3-cd) pyrene	0.0022
93	Isophorone	4.2
94	<i>Naphthalene</i>	<i>17</i>
95	Nitrobenzene	8.5
96	N-Nitrosodimethylamine	0.00035
97	N-Nitrosodi-N-propylamine	0.0025
98	N-Nitrosodiphenylamine	2.5
99	Phenanthrene	--

	CONSTITUENT	µg/L
100	Pyrene	480
101	<i>1,2,4 -Trichlorobenzene</i>	<i>5</i>
102	Aldrin	0.00007
103	BHC Alpha	0.0020
104	BHC Beta	0.007
105	BHC Gamma	0.010
106	BHC Delta	--
107	Chlordane	0.00029
108	4,4-DDT	0.0003
109	4,4-DDE	0.0003
110	4,4-DDD	0.00042
111	Dieldrin	0.00007
112	Endosulfan Alpha	0.028
113	Endosulfan Beta	0.028
114	Endosulfan Sulfate	55
115	Endrin	0.018
116	Endrin Aldehyde	0.38
117	Heptachlor	0.00011
118	Heptachlor Epoxide	0.00005
119	PCB 1016	0.000085
120	PCB 1221	0.000085
125	PCB 1260	0.000085
126	Toxaphene	0.00037

Notes:

1. For constituents not shown italicized, the values shown in the Table are fifty percent of the most stringent applicable receiving water objectives (freshwater or human health (consumption of water and organisms) as specified for that pollutant in 40 CFR 131.38⁶).
2. For constituents shown bold and italicized, the values shown in the Table are based on the California Department of Health Services maximum contaminant levels (MCLs) or Notification Level. Notification Level based trigger is underlined.
3. For hardness dependent metals, the hardness value used is 154 mg/L and for pentachlorophenol, the pH value used is 7.5 standard units.

⁶ See Federal Register/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations.